



On the Modeling and Simulation of SLM and SLS for Metal and Polymer Powders: A Review

Emmanouil L. Papazoglou¹ · Nikolaos E. Karkalos¹ · Panagiotis Karmiris-Obratański^{1,2} · Angelos P. Markopoulos¹

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Abstract

Additive Manufacturing concentrates the attention, not only of the research and academic community, but of the industry as well. Selective Laser Melting (SLM) and Selective Laser Sintering (SLS) are among the broadest employed methods in AM, since they can treat almost all types of materials. Along with the extensive experimental research that is carried out regarding SLS and SLM, modeling and simulation are powerful tools allowing better and more in depth understanding of the processes. Nevertheless, there is no general framework in modeling, but mainly studies and proposed modeling approaches. The current paper reviews modeling methods and techniques that in literature are presented for the simulation of SLM and SLS. Besides the Finite Element Method, which is the most common method used, other numerical methods like Discrete Element Method, Smoothed Particles Hydrodynamics and Molecular Dynamics have been overviewed as well. The heat transfer and fluid dynamics models consist the main core of every simulation, while other sub-models are integrated to estimate parameters like residual stresses, part deformation, material microstructure, or crystallization. The main scope of the current paper is to provide a comprehensive and detailed review on the modeling and simulation of SLS/SLM and to inform the reader concerning the different modeling strategies.

1 Introduction

Additive Manufacturing (AM), which is also labeled as 3D printing, Rapid Prototyping (RP), Layered Manufacturing (LM), or Solid Freeform Fabrication (SFF), is a novel manufacturing process, with basic principle that 3D designs can be built directly from a Computer Aided Design (CAD) file without using any part—specific tools and/or dies [1]. Conceptually, in AM, parts and components are made by adding and stacking layers of material, with each layer being a thin cross section of the final part [2]. Moreover, due to near net shape processing, AM creates less waste than conventional machining, while due to the absence of design restrictions, optimization is feasible, resulting to lighter and stronger parts. Utilizing the unprecedented manufacturing

versatility of AM and its inherent advantages over traditional machining processes, 3D printing finds extensive use in the modern industrial environment [1]. Aerospace, automotive, oil and gas industries have already incorporated and invested on AM technology, while 3D printing finds numerous applications in the medical and dental field, as well in materials and structures engineering [3, 4].

In the recent years, researchers have come up with many alternative AM processes, which can handle all types of materials, utilize different kinds of energy sources and ways to consolidate the stacked layers. As a result, the sorting of AM processes can be done according to the material type, the material state, the energy source or the consolidation method. Commonly the AM processes are classified in liquid based, solid based and powder based according the state of the raw material [5, 6]. Powder and liquid based methods are considered as the most promising, regarding their versatility and capabilities, as they include and can be applied in a broad spectrum of materials. Selective Laser Melting (SLM) and Selective Laser Sintering (SLS) belong to the powder based processes, being currently two of the most rapidly developing technologies.

✉ Angelos P. Markopoulos
amark@mail.ntua.gr

¹ School of Mechanical Engineering, Department of Manufacturing Technology, National Technical University of Athens, Athens, Greece

² Faculty of Mechanical Engineering and Robotics - Department of Manufacturing Systems, AGH University of Science and Technology, Cracow, Poland

1.1 Selective Laser Sintering (SLS) and Selective Laser Melting (SLM)

SLS and SLM are both powder-based AM processes, hence, they have many similarities and share common basic principles. Nevertheless, it is important to be identified as distinct and different processes, as they have unique and special features, characteristics, capabilities and limitations.

SLS has already about three decades of life, since 1989, and conceptually, it uses a high-power laser to fuse small particles of raw material, in order for a 3D solid object to be build. The powder bed is heated just below material's melting point, aiming to minimizing thermal distortions, while the concurrent fusion to the previous layer can be facilitated. From the absorbed laser beam energy, the material is locally and partially melted, forming the contour of the part's layer, while the unsintered powder remains in place to support the structure. As the laser beam moves, material is solidified due to heat transfer by convection, conduction and radiation. After the layer's solidification, the powder bed is lowering by the amount of one layer thickness, with the emerged empty space on the top of the powder bed being filled with new powder material, and the sintering/build process starts again [5]. In SLS various materials can be used, like metals, polymers, ceramics and composite materials, while different binding/sintering mechanisms are utilized. In Solid State Sintering (SSS) the thermal process is taking place in temperatures below the material's melting point, with diffusion being the dominant and most important sintering mechanism. SSS is, however, a slow process and the powder preheating is required. In contrast with SSS, in Liquid Phase Sintering (LPS) a structural material remains in solid state, while a binder material is liquefied; the liquid and solid phases may come from the same powder material. LPS includes many different kinds of technologies [7]. Since sintered parts, especially polymers, often appear intense porosity, a post-treatment may be needed [8].

Thus, driven by the necessity to produce near full dense objects by using AM and in order for post-processing cycles to be avoided, SLM was developed. In SLM the raw powder material is fully melted by the laser beam, resulting to local (selective) melt pools, which in turn result to fully dense parts (> 99% density) after the re-solidification. Although polymers can fully melt by a laser beam, in SLM, metal powders, e.g. steel, Ti or Al alloys, are commonly used. The capability to manufacture high density [9] and strength parts, with complex shapes and geometries, the negligible waste of material and the ability of processing a wide variety of metals and alloys with no need of distinct binders or melt phases, are some of the unique advantages of SLM. Nevertheless, SLM as a newly developed process is facing challenges that have to be overcome in order for SLM to become

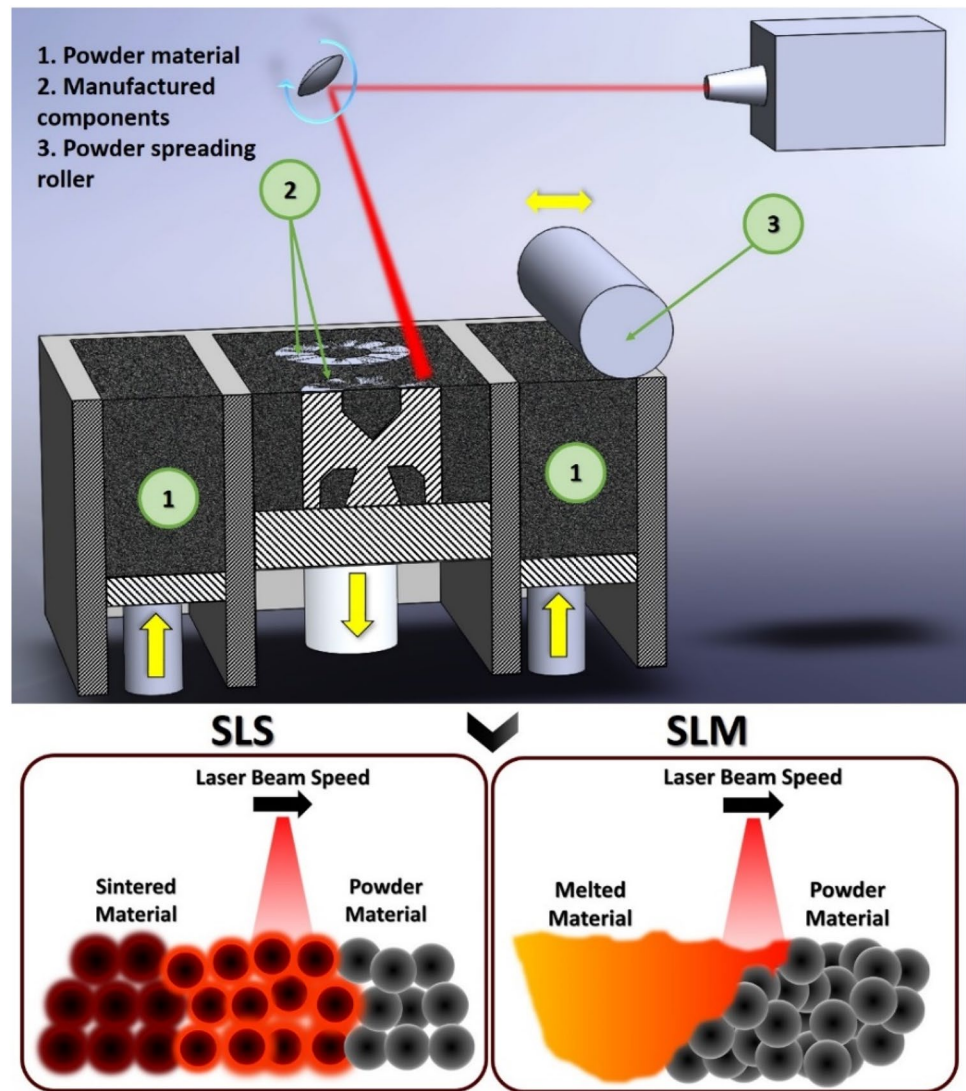
a competitive manufacturing alternative process. SLM suffers from melt pool instabilities, leading to parts' defects and imperfections, like low quality down-facing surfaces, increased upper-surface roughness and the risk of internal porosity. Additionally, high gradients in temperature and pressure increase the risk of delamination and/or distortion due to high thermal stresses that are developed [5, 8, 10]. In Fig. 1, SLS and SLM processes are schematically presented.

1.2 Need for Modeling and Simulation

Although there is a great deal of excitement in scientific and engineering community regarding AM, these processes are inherently complicated, with multi-physics phenomena taking place. Researchers have identified over 130 parameters that are affecting the process and the final quality of the manufactured part. As an example of the complexity of the process is the interaction of the laser beam, namely electromagnetic energy, with the material of the powder bed, which at first is in solid powder state but subsequently is turned into liquid and even in gas. This interaction is related with the energy absorption from the material, thus the thermal equilibrium and the developed temperature profile. Additionally, the laser beam's type, power, spatial distribution and speed, the powder's thermophysical properties, and process' parameters, like the preheat temperature and the build chamber environment, are only a few factors that affect the SLS and SLM and have to be studied carefully and in depth [11, 12]. Moreover, material transformation from solid phase to liquid and vice versa, introduce the concept of material's temperature history, which may differ from region to region. Furthermore, the cooling rate determines the particle size and the metallurgical composition of the microstructure, and hence, component's mechanical properties such as stiffness, elasticity and yield strength. Furthermore, the expansion and contraction of both the newly added material and the existing layers may lead in high residual stresses, resulting even to the component's permanent warping. All the aforementioned parameters have to be taken into consideration in the production of every new part, as the final product has to be within certain tolerance limits, and with sufficiently reliable material properties for the end use [13].

Aiming at an efficient manufacturing by employing SLS and SLM methods, modeling and simulation are powerful tools in the effort to better and fully understand those processes and to optimize them as well. Nevertheless, modeling of SLM faces some significant challenges on its own. At first, the material addition has to be simulated and modeled in an adequate accuracy. At the same time, the model has inescapable non-linearities, since most of the considered parameters are temperature dependent, like the material properties, the radiation losses, the material plasticity and the occurred deformation. In addition, the model and the corresponding

Fig. 1 Schematic presentation of SLS and SLM processes



computations have to be discretized both spatially and temporally, requiring significant computational resources. Thus, a deep understanding of computational methods is required, for accurate results to be produced during a reasonable time frame. Thus, the challenges in modeling and simulation of SLS and SLM can be summarized in the following:

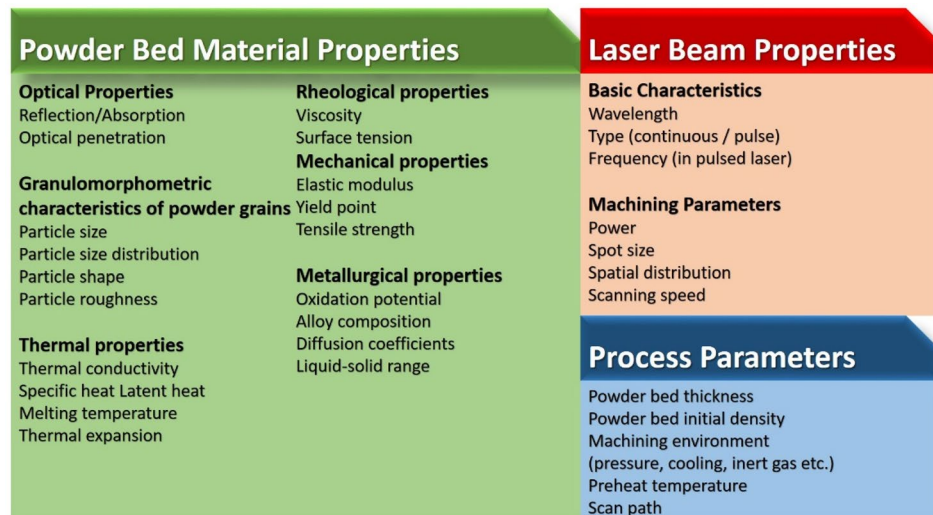
- Modeling the material addition process
- Realistic modeling of the heat source
- Accurate accounting of thermal losses during the process
- Coupling the heat transfer analysis with fluid dynamics in a multi-physics thermo-hydraulic simulation
- Accurate modeling of elasto-plastic stresses and strains
- Coupling the thermo-mechanical behavior in a manageable manner
- Taking into consideration the temperature dependence of material properties, from room temperature through melting point and even higher
- Modeling the phenomena that affect material properties, i.e. changes in microstructure, phase transformation, anisotropy development, etc.

In Fig. 2 the most important parameters that have to be modeled and/or defined for an accurate simulation of SLS and SLM are presented.

1.3 On the Computational Power and Time

Computational cost is an important issue in every simulation and especially in the most demanding cases, as the SLM and SLS simulations. As it was aforementioned, these processes involve a large variety of interacting variables that it is impossible to study them at the same time [14]. These parameters can affect the computational cost of the simulation and thus it is crucial to mention them in brief. One of the most important features of a computational model is the

Fig. 2 Major parameters in modeling SLS and SLM



mesh, which is directly related to the degrees of freedom of the model and subsequently to the computational time and memory requirements of the simulation which is a function of the total degrees of freedom. As it is not efficient to create an extremely fine mesh in the whole model, the mesh should be refined in the areas where the laser irradiation takes place and the solidification takes place, whereas far from this region the mesh should become gradually coarser. Moreover, another efficient technique discussed by several authors is the use of an adaptive remeshing technique which focused on the most important areas for each stage of the simulation and leads to a considerable reduction of both computational time and memory requirements [15–18]. Apart from the adaptive remeshing approach, multi-scale approaches are also possible, with micro-scale results transferred to a different scale in order to improve accuracy [19]. Another simplification usually conducted in the relevant literature is the use of symmetry boundary conditions in order to reduce the size of the model. Furthermore, due to the high computational cost, only the creation of single tracks is usually modeled in FE models of SLM and SLS processes and not the manufacturing of entire 3D parts [16].

The use of different physics models is another crucial issue in SLM and SLS simulations, as the computational cost of a simple thermal model is considerably increased when fluid flow of molten material, mechanical behavior, microstructure and the discrete nature of powder are taken into consideration [15]. Thus, it is essential that the researchers choose the desired level of complexity in their model according to the requirements of their study. For example, the discrete nature of the powder should be taken into consideration when the study focuses on the sintering or melting process of the powder particles but on the other hand, it can be simplified with a continuum model in other cases when quantities of a high length scale are to be calculated e.g.

the melt pool dimensions. Moreover, the use of temperature dependent properties renders the simulation non-linear and increases the computational cost. It should be noted that simplifications are possible in all the aforementioned models. For example, an equivalent heat conduction coefficient, taking also into consideration melt pool flow effects can increase the accuracy of a simplified model which does not include Marangoni and buoyancy effects [18]. Similarly, the effect of material microstructure can be either included via a material model or more directly with a mesoscale model, at an increased computational cost [20].

Finally, although it is rarely mentioned in the relevant literature, typical simulation times of SLM models require several hours up to weeks in order to be completed with current hardware, but can be considerably reduced when the employed software can be efficiently parallelized or when GPU acceleration is possible [16, 21].

2 Thermal Model

The heat transfer model of an SLM system has to take into consideration both input and output heat fluxes. In general, the major, and in most cases the only, input heat source is the laser beam, while the output fluxes include convection and radiation losses, along with any heat flux occurred, due to ablated material. The conduction governing equation is [22]:

$$\rho C_p \frac{\partial T}{\partial t} - \nabla \cdot (k \nabla T) = \dot{Q} \quad (1)$$

where ρ the material's density in [kg/m³], C_p the specific heat in [W/m²K], k the thermal conductivity [W/mK] and $\dot{Q}$ the heat rate per unit volume [W/m³s]. This basic heat transfer equation is used in almost every SLM model, with the proper changes and variations according to the model

assumptions [23–25]. The more interesting and notable part of this equation is the determination of every term, in a realistic and computationally efficient way.

2.1 Laser Beam Heat Source

Although the laser beam that is normally utilized in SLM has a bell-shape Gaussian spatial distribution, many different modeling methods have been developed to simulate it. A first differentiation can be done according to whether the laser beam is defined as a surface or volumetric heat source. In the first and simpler case, the power intensity, by using the Gaussian distribution formula, can be expressed as [26]:

$$Q_{LB}(x, y) = \frac{P\eta}{2\pi\sigma^2} \cdot e^{-\frac{1}{2}\left(\frac{x^2+y^2}{\sigma^2}\right)} \quad (2)$$

with Q_{LB} the power intensity in [W/m²], P the laser beam nominal power in [W], η the absorption coefficient, and σ the standard deviation.

The standard deviation σ is defined proportionally to the radius, where beam's intensity is reduced by a factor of e^2 , in comparison to the intensity at the center of the beam, thus, $\sigma = r_{LB}/2$ with r_{LB} the beam's waist radius [26–34]. This approach, of a symmetrical heat source, is very close to the real laser beam, especially in processes where Yb-fiber lasers were used, which provide an almost perfect Gaussian-shaped beam [10].

The symmetrical heat source is a realistic approach, especially in modeling processes where Yb-fiber lasers are used, which provide an almost perfect Gaussian—shaped beam. Nevertheless, ellipsoidal distribution heat sources have been adopted as well. The basic Gaussian distribution formula is utilized, but it is properly modified, in order to incorporate the width and length of the ellipsoid. In this case, the power intensity is described by the following equation [13]:

$$Q_{LB}(x, y, t) = \frac{3P\eta}{\pi x_o y_o} \cdot e^{-\left(\frac{3x^2}{x_o^2} + \frac{3y^2}{y_o^2}\right)} \quad (3)$$

with x_o and y_o the heat source length and width in [m] respectively.

Except the 2D surface models, volumetric heat sources have been adopted as well. It is known that the electromagnetic energy of the laser beam is absorbed by a layer of material and not only by its surface. Nevertheless, the aforementioned layer is in the order of some nm, significantly smaller than the grains size and the powder bed thickness [35]. Therefore, the volumetric heat source is not referred to the absorption of the laser beam by the material, but rather to the multi scatterings of the laser beam across the powder bed volume. The laser beam's energy absorption and the corresponding heat source model is a really interesting topic

and for its realistic simulation different volumetric distribution have been proposed.

In the most simple model, a cylindrical heat source with a uniform energy distribution is proposed and the power density is calculated according to the following equation [36, 37]:

$$Q_{LB}(x, y, z) = \frac{\eta P}{\eta_{OPD} S_{LB} z_{OPD}} \quad (4)$$

with S_{LB} the area of the laser beam spot in [m²], z_{OPD} the Optical Penetration Depth (OPD) in [m] and η_{OPD} the correction factor for the assumed OPD. The OPD can be determined as equal to the powder bed thickness, when there is lack of additional experimental or computational data [36].

In a step towards a more accurate model, a three-dimensional Gaussian intensity distribution has been suggested. This heat source has a semi-spherical Gaussian distribution of power density and is mathematically expressed as [36, 37]:

$$Q_{LB}(x, y, z) = \frac{2^{\frac{5}{2}} \eta P}{\pi^{\frac{3}{2}} r_{LB}^3} \cdot e^{-2\left(\frac{x^2+y^2+z^2}{r_{LB}^2}\right)} \quad (5)$$

In another variant of the volumetric heat source, a model with linear decay along the OPD has been employed. In this case, the power density is expressed as [36, 38]:

$$Q_{LB}(x, y, z) = \frac{2P\eta}{\pi r_{LB}^2} \cdot e^{-2\left(\frac{x^2+y^2}{r_{LB}^2}\right)} \frac{2}{z_{OPD}} \left(1 - \frac{z}{z_{OPD}}\right) \quad (6)$$

Similarly to the previous model, a volumetric heat source with exponential decay along the OPD, has been proposed as well, and described by Eq. 7 [36, 39]:

$$Q_{LB}(x, y, z) = \frac{2P\eta}{\pi r_{LB}^2} \cdot e^{-2\left(\frac{x^2+y^2}{r_{LB}^2}\right)} \frac{1}{z_{OPD}} e^{-\frac{|z|}{z_{OPD}}} \quad (7)$$

In a more sophisticated approach, a semi-ellipsoidal distribution model has been suggested. This is well known as "Goldak" heat source model, and it has been originally developed and used in the modeling of welding. There are two "versions" of the Goldak model, the single ellipsoidal and the double ellipsoidal model. In the second one, the front part of the ellipsoid could be different from the rear part, thus, the model becomes more versatile, in order to better fit with some experimental results. The single and double ellipsoidal model are mathematically described by Eqs. 8 and 9 respectively [36, 40].

$$Q_{LB}(x, y, z) = \frac{2^{\frac{5}{2}} \eta P}{\pi^{\frac{3}{2}} x_{el} y_{el} z_{el}} \cdot e^{-2\left(\frac{x^2}{x_{el}^2} + \frac{y^2}{y_{el}^2} + \frac{z^2}{z_{el}^2}\right)} \quad (8)$$

with x_{el} , y_{el} , z_{el} the ellipsoid semi-axes.

$$Q_{LBf}(x, y, z) = \frac{2^{\frac{5}{2}} \eta f_f P}{\pi^{\frac{3}{2}} x_{f,el} y_{el} z_{el}} \cdot e^{-2 \left(\frac{x^2}{x_{f,el}^2} + \frac{y^2}{y_{el}^2} + \frac{z^2}{z_{el}^2} \right)} \quad Q_{LBf}(x, y, z) = \frac{2^{\frac{5}{2}} \eta f_r P}{\pi^{\frac{3}{2}} x_{r,el} y_{el} z_{el}} \cdot e^{-2 \left(\frac{x^2}{x_{r,el}^2} + \frac{y^2}{y_{el}^2} + \frac{z^2}{z_{el}^2} \right)} \quad (9)$$

with Q_{LBf} and Q_{LBf} the power density of the front and rear ellipsoidal distribution respectively, f_f, f_r the power fraction of front and rear parts respectively, $x_{f,el}$ the front ellipsoid's semi-axis, and $x_{r,el}$ the rear part semi-axis. Because of the energy equilibrium, $f_f + f_r = 2$ [36, 40].

Although the Goldak heat source model has been extensively used in SLM modeling, there is an inherent difficulty in its use, as there are six constants that have to be defined, namely $f_f, f_r, x_{f,el}, x_{r,el}, y_{el}$ and z_{el} . The calibration of the model is commonly done by fitting it with experimental results [41], nevertheless, an assumption has been proposed in order to limit the unknown constants, correlating the energy fraction of each part with part's semi-axis. This constrain is mathematically expressed as [42, 43]:

$$\frac{f_f}{x_{f,el}} = \frac{f_r}{x_{r,el}} \quad (10)$$

Although the Goldak model seems to be more versatile than other suggested ones, it has the fundamental disadvantage to need a calibration, as it is based on six unknown constants. Hence, modeling the SLM process using a Goldak heat source, strongly depends on the availability of experimental results, not only for model's validation, but to define its important parameters, as well. Therefore, such a model is highly case-sensitive, with limited universal applicability and generality.

Finally, in a more plain approach, the laser beam is modeled using a point heat source. In this way, there is an analytical solution for the temperature profile [44], rendering the calculations simpler and far less time consuming, with limited needs in computational power. Even though such a model is not as realistic as the previous ones, taking into consideration its pros according to simplicity and solving speed, it has been adopted by many researchers [43, 45, 46].

All the aforementioned models incorporate the absorption coefficient " η ", which defines the percentage of the laser beam's nominal power that is absorbed by the material during the SLM process. The calculation and determination of the absorption coefficient is a very interesting and important scientific topic. Generally speaking, three major approaches are followed in determination of absorption coefficient: either a semi-empirical coefficient is adopted according to the literature review, or a coefficient that results the best fit between experimental and simulation results, or the coefficient is calculated based on principles of physics and

the corresponding relations. Its precise definition is a very

"demanding" task since numerous variables are involved. The interaction between the laser beam's electromagnetic wave and powder bed's material is a complicated system, since the beam is not projected on a flat surface, but on grains of material, the optical properties are temperature dependent [41], while during the process, material's solid and liquid phase may coexist. Primarily, the laser beam wavelength, materials' optical properties, and the granulomorphometric characteristics of the powder material are some major parameters. Additionally, the laser beam's intensity, speed, spot size and the keyhole formation affect the energy absorption [47, 48].

Experiment conduction is surely an efficient and accurate method in order to determine the absorption coefficient, nevertheless, it has significant requirements for special equipment, while results have limited applicability since they refer to specific experimental conditions and parameters. Thus, theoretically, a significant number of experiments have to be carried out, for different combinations of materials and laser beam types, a time consuming and unbearably expensive procedure. As a result, alternative strategies and practices have been adopted, like the implementation of semi-empirical coefficients, or their calculation by using analytical and/or computational models.

In the work of Loh et al. [23], the absorptivity coefficient of Aluminum Alloy 6061 for Yb:YAG fiber laser was defined as 0.48, and Qi et al. [49] in their model considered 47% absorptivity for Al7050. In the study of Doubenskaia et al. [50] and Arrizubieta et al. [51] for Continuous Wave (CW) laser at 1.07 μm , the absorption coefficient of stainless steel 304 L was taken as 0.3. An interesting analysis about the absorptivity of 316L stainless steel at 1.075 μm wavelength was presented by Gunenthiram et al. [52], concluding that the absorbance mainly takes place while the material is in liquid state and is comprised between 58 and 62%. Nevertheless, Conti et al. [53] assumed for 316L a 30% absorbance, while in the work of Chen et al. [54] a 40% coefficient for 316L is referred. Lee et al. [31] used a 0.38 coefficient regarding the absorbance of Yb-fiber laser by IN718 alloy, while Kundakc et al. [55] considered for Inconel 625 a 0.37 (almost similar) absorption coefficient. In modeling of Ti-6Al-4 V, different coefficients have been employed, namely, 0.77 by Mirkoochi et al. [43], 0.45 by Yang et al. [25], a little lower, at 0.4 by Zhang et al. [56], 0.3 was adopted in the study of Yang et al. [57], Zhuang et al. [58] and Ansari et al. [59], 0.24 in the work of Panwisawas [60] and 0.2 by

Ye et al. [34]. For AISi10Mg at 1.07 μm wavelength Liu et al. [39] used an 60% absorbance coefficient, while Panda et al. [26] proposed a 90% absorbance factor for the same material. Finally, Wu et al. [61] adopted an extremely low value of 10% for H13 stainless steel. In Table 1 a synopsis of the absorbance coefficient for the most common materials in SLM is presented.

This significant deviation in the absorption coefficient values can be mainly attributed to the different modeling assumptions and the simplifications that each study adopts. For example, in modeling the 3D printing of Ti–6Al–4 V, the absorption coefficient ranges from 0.2 up to 0.77. The lower values are in line with and have been adopted from studies regarding the laser welding of titanium alloys, the middle values (i.e., 0.4 and 0.45) are close and refer to the absorption coefficient of Titanium based on its refractive

Thus, in order to avoid the divergence in the implemented absorption coefficients, computational models, based on physics principles, have been proposed and used in SLM modeling. One common and efficient technique is the estimation of the absorption coefficient following the Ray Tracing Model (RTM). Conceptually, in the RTM, the laser beam is divided into rays with certain size, direction and amount of power, which incident on the surface of the powder bed. A portion of the energy is absorbed, while the rest is reflected, and by employing the reflection law, the direction of reflected ray and the absorbed power can be calculated. Thus, the total ray absorbance due to multiple scattering along the powder bed can be estimated [63]. In the core of the RTM are the Fresnel formulas, which provide the absorptivity, S polarization and P polarization at an incident angle ϑ , and are mathematically expressed as [64]:

$$\alpha_s(\vartheta) = 1 - \left| \frac{\cos \vartheta - (n^2 - \sin^2 \vartheta)^{1/2}}{\cos \vartheta + (n^2 - \sin^2 \vartheta)^{1/2}} \right|^2 \quad \alpha_p(\vartheta) = 1 - \left| \frac{n^2 \cos \vartheta - (n^2 - \sin^2 \vartheta)^{1/2}}{n^2 \cos \vartheta + (n^2 - \sin^2 \vartheta)^{1/2}} \right|^2 \quad (11)$$

indexes and the Fresnel equations, while the 0.77 value is straight coming from studies regarding the SLM. It has to be pointed out that none of these values on its own is correct or incorrect, since model's basic assumption have always to be carefully considered. Namely, when due to the machining parameters high power densities are arise, and during the process plasma shielding effect is occurred, then, the ultimate absorption coefficient of the powder is limited in lower values. Based on a pure theoretical analysis absorption values according to material's refractive indexes are again justified to be used, but now a weighting coefficient has also to be considered, since the process is not ideal and flawless, and complex phenomena are taking place. Finally, if the multiple scattering effect of the laser beam and the absorption by metal's liquid phase are considered, then a much higher initial absorption coefficient can be utilized, but now, phenomena like the material ablation and the plasma shielding have also to be modeled and taken into the account. Hence, during the modeling and the determination of the boundary conditions and coefficients it is important and necessary a clear and detailed state of model's major and primarily assumptions and simplifications.

where, α_s and α_p refer to the absorption to S-polarized and P-polarized light, respectively, and n is the relative refractive index of the powder material. RTM has been employed by several researchers, as it provides the necessary versatility to calculate the laser beam absorption more accurately, taking into consideration the multiple scatterings and material's optical properties [65, 66]. Moreover, utilizing the RTM, calculations about the absorbance by using different powder packing models can be easily performed [67, 68]. In the work of Tran et al. [69] the absorptivity profile along the powder bed thickness is unified with a 2D Gaussian heat source, resulting to a hybrid volumetric heat source. With that elegant formula, not only the total absorption can be estimated, but at the same time, the absorptivity by each sub-layer can be calculated as well, providing a more in depth and versatile analysis. The mathematical expression is:

$$Q_{LB}(x, y, z) = \frac{2P}{\pi r_{LB}^2} \cdot e^{-2\left(\frac{x^2+y^2}{r_{LB}^2}\right)} \frac{da}{dz} \quad (12)$$

with da/dz the absorptivity profile along the OPD.

Table 1 Absorbance coefficients of the most common materials in SLM

Al6001	Al7050	IN718	IN625	H13 SS	304L SS	AISi10Mg	316L SS	Ti–6Al–4 V
0.48 [23]	0.47[49]	0.38 [31]	0.37 [55]	0.1[61]	0.3 [50, 51]	0.60 [39] 0.90 [26]	0.58 – 0.62 [52] [62] 0.3 [53] 0.4 [54]	0.77 [43] 0.45 [25] 0.4 [56] 0.3 [57–59] 0.24 [60] 0.2 [34]

In a modification of the RTM, Zhou et al. [70] employed a packing algorithm to generate 3D randomly packed structures that are composed of spherical particles with the same or different sizes. Subsequently, assuming diffusively scattering on the powder, the Monte Carlo sampling technique was applied to generate random reflection direction. Since the assumption of gray surface has been adopted, its reflectivity is computed based on the emissivity.

Gusarov et al. [71] proposed a model where the laser radiation transfer in metallic powders is calculated according to the radiation transfer equation. According to that model, the laser beam penetrates into a depth of a powder-bed, which has z_{PB} thickness, and η_{ext} extinction coefficient. The model is mathematically described as [36, 72]:

$$Q_{LB}(x, y, z) = \frac{2P}{\pi r_{LB}^2} \cdot e^{-2\left(\frac{z^2+y^2}{r_{LB}^2}\right)} \left(-\eta_{ext} \frac{dq_R}{d\xi} \right) \quad (13)$$

where $\xi = \eta_{ext} \cdot z$ is the dimensionless local depth coordinate, and q_R is the dimensionless form of net radiative energy flux density and is described as:

$$q_R = \frac{\gamma a_s}{(4\gamma - 3)D} \left\{ \begin{aligned} &(1 - \gamma^2) e^{-\lambda} \left[(1 - a_{hem}) e^{-2\alpha_s \xi} + (1 + a_{hem}) e^{2\alpha_{hem} \xi} \right] - \\ &-(3 + \gamma e^{-2\lambda}) \left[(1 + \alpha_{hem} - \gamma(1 - \alpha_{hem})) e^{2\alpha_{hem}(\lambda - \xi)} + \right. \\ &\left. + (1 - \alpha_{hem} - \gamma(1 + \alpha_{hem})) e^{2\alpha_{hem}(\xi - \lambda)} \right] \end{aligned} \right\} - \frac{3(1 - \gamma)(e^{-\xi} - \gamma e^{\xi - 2\lambda})}{4\gamma - 3} \quad (14)$$

where $\lambda = \eta_{ext} \cdot z_{PB}$ refers to the optical thickness for the powder-bed, $a_{hem} = \sqrt{1 - \gamma}$, γ is the hemispherical reflectivity in the dense form, and D and η_{ext} are described as:

$$D = (1 - a_s) [1 - a_{hem} - \gamma(1 + \alpha_{hem}) e^{-2\alpha_{hem} \lambda}] - (1 + a_{hem}) [1 + a_{hem} - \gamma(1 - \alpha_{hem}) e^{2\alpha_{hem} \lambda}] \quad (15)$$

$$\eta_{ext} = \frac{S_p}{4} \quad (16)$$

where S_p is the specific powder surface per unit pore volume, γ is 0.7 [36], S_p is regarded as π/r_{GRAIN} and r_{GRAIN} is the average powder's grain radius.

Finally, to estimate the absorption coefficient, some other models have been proposed and presented as well. Chen et al. [73] in their study developed a model based on the Beer-Lambert law about the attenuation of light according to the properties of the material through which the light is traveling. Thompson et al. [30] proposed a relation that utilizes the temperature-dependent electrical resistivity of the material through Bramson's equation or the Hagens-Rubens relationship and expressed as:

$$a(T) = \sqrt{8\varepsilon_o \omega \rho_e(T)} \quad (17)$$

where $a(T)$ is the absorption coefficient, ε_o is the permittivity of free space, ω is the angular frequency of the laser

irradiation (rad/s) and $\rho_e(T)$ is the material's temperature-variant electrical resistivity.

At this point, it has to be accentuated that the aforementioned methods to calculate the absorption coefficient are related to the interaction between a laser beam's electromagnetic energy and powder bed's solid material. However, during SLM, at some point, the laser beam may project on molten material, thus, another model has to be implemented for the most accurate and realistic absorption coefficient calculation. For temperatures above the material's melting point, temperature dependent absorption coefficients can be calculated by utilizing the concept of electron—phonon optical conductivity, and the Drude theory of optical properties of metals. Siegel [74] has presented a very interesting and useful study regarding the optical reflectivity of liquid metals at their melting temperature.

2.2 Heat Losses of Convection, Radiation, and Material Ablation

During SLM, heat transfer due to convection and radiation

occurs, which can be described as heat loss. The convection can be natural or forced, according to the conditions inside the SLM chamber and the movement of the shielding

gas. In general, heat losses due to convection are low and in some studies they are neglected for the sake of economy in computational time and power [23, 75, 76]. Nevertheless, in a detailed analysis they have to be taken into consideration and contribute to the system power equilibrium. Mathematically, the heat transfer due to convection is described by Eq. 18 [77]:

$$q_{CONV} = h_{CONV}(T - T_\infty) \quad (18)$$

with q_{CONV} the heat losses due to convection in [W/m²], h_{CONV} the convective heat transfer coefficient in [W/m²K], and T_∞ the environment temperature in [K].

The calculation of heat transfer due to convection, presupposes the convective heat transfer coefficient estimation. The most common modeling method is the adoption of a constant coefficient according to the powder bed material. Actually, this coefficient is temperature dependent and its accurate calculation is quite difficult, nevertheless, the use of a

semi-empirical coefficient, in a realistic order of magnitude, consists a practical modeling approach. For various materials, various coefficients have been proposed; sometimes for the same material, different coefficients can be found in the relevant literature. In Table 2 coefficients coming from the literature review are listed.

Again, it is observed a deviation in convective coefficients, a difference that may become even an order of magnitude. This divergence is due to the modeling assumptions and the adopted simplification hypotheses. In some studies, the convective coefficient is defined based on a free convection over a solid plate, while, in other models, there is distinction between forced and free convection, and between the heat transfer on the wall areas and the substrate areas. Moreover, different approach and values are justified in case where a mean coefficient is employed, or a temperature dependent one. Hence, it has to be clearly stated and reasonably justified any value that is used, along with the respective assumptions, in order each model to retain its scientific background and robustness.

In an effort to estimate the convection coefficient more accurately, and taking into consideration not only the powder bed material, but the environment of the SLM chamber, the calculation in basis of the Nusselt number has been proposed by Zhuang et al. [58]. The convection coefficient can be calculated according to Eq. 19 [77]:

$$h_{CONV}(T) = \frac{Nu(T) \cdot k_{env}(T)}{L} \quad (19)$$

where L is the characteristic length of the specimen, Nu is the Nusselt number and k_{env} is the thermal conductivity of the fluid in the atmosphere in [W/mK]. Considering the formed molten pool as a hot horizontal plate, Nu can be calculated as [22, 58]:

$$Nu(T') = 0.54(Gr(T') \cdot Pr(T'))^{\frac{1}{4}} \text{ with } T' = \frac{T_{\infty} + T}{2} [K] \quad (20)$$

with Gr the Grashof number and Pr the Prandtl number. Equation 20 can be implemented only if Rayleigh number ($Ra_L = GrPr$) is $10^4 \leq Ra_L \leq 10^7$. Grashof and Prandtl numbers are calculated according to Eqs. 20 and 21, respectively.

$$Gr = \frac{g \rho_{env} \beta_{env} (T - T_{\infty}) L^3}{\mu_{env}^2} \quad (21)$$

$$Pr = \frac{C_{p_{env}} \mu_{env}}{k_{env}} \quad (22)$$

where g is the acceleration of gravity in [m/s²], ρ_{env} the density in [kg/m³], β_{env} the volumetric expansivity in [K⁻¹], μ_{env} the viscosity in [kg/m·s], $C_{p_{env}}$ the specific heat in [J/kgK], and L the characteristic length.

The heat losses due to radiation are mathematically expressed as [77]:

$$q_R = k_B \epsilon (T^4 - T_{\infty}^4) \quad (23)$$

with q_R the heat losses due to radiation in [W/m²], ϵ is the emissivity coefficient, and k_B the Stefan–Boltzmann constant, 5.67×10^{-8} [W/m²K⁴].

Again, the accurate calculation of heat transfer due to radiation depends on the estimation of the emissivity coefficient. By the most common practice, the emissivity coefficient is adopted according to the powder bed material, based on the literature available data. However, at this point, two important remarks have to be made. Firstly, the radiation surface is not a solid material surface, but the summation of powder's granules surfaces. Secondly, the material, surely will not remain solid throughout the whole process, on the contrary, a molten pool of material is formed. Thus, these two significant details have to be considered, working towards an accurate and realistic modeling. The deviation between emissivity of solid and liquid phases can be handled by incorporating a different emissivity coefficient for temperatures above the material's melting point. In the bibliography there are studies about the emissivity of melted metals [79, 80], and so, for temperatures higher than material's melting point, a temperature dependent emissivity coefficient can be adopted.

To deal with the emissivity of the powder bed, the Zehner–Schlunder model [81] can be mentioned and the study of Sih and Barlow [82]. The emissivity of a powder bed surface can be calculated by knowing only the bed porosity and the emissivity of its solid phase, according to the following relation:

Table 2 Convective heat transfer coefficient

AISI 304	AISi10Mg	316L SS	IN718	Ti–6Al–4 V
20 [W/m ² K] [51]	80 [W/m ² K] [78]	50 [W/m ² K] [62]	80 [W/m ² K] [31]	5 [W/m ² K] [25]
				≈10 [W/m ² K] [24]
				12.7 [W/m ² K] [41]
				100 [W/m ² K] [32]

$$\varepsilon_{T < T_m} = A_H \varepsilon_H + (1 - A_H) \varepsilon_S \quad (24)$$

where ε_H is the emissivity of cavities (holes) of the powder bed, ε_S is the emissivity of solid particles in the powder bed and A_H is the area fraction of the surface that is occupied by the radiation emitting cavities. The area fraction can be calculated as:

$$A_H = \frac{0.908 \cdot \varphi^2}{1.908 \cdot \varphi^2 - 2 \cdot \varphi + 1} \quad (25)$$

with φ porosity of the powder bed. The emissivity of the cavities can be estimated as a function of solid particle emissivity and bed porosity as follows:

$$\varepsilon_H = \frac{\varepsilon_S \left[2 + 3.082 \left(\frac{1-\varphi}{\varphi} \right)^2 \right]}{\varepsilon_S \left[1 + 3.082 \left(\frac{1-\varphi}{\varphi} \right)^2 \right] + 1} \quad (26)$$

Finally, the heat losses due to ablation of the material can be determined in respect to ablated material mass, and thus, the heat flux can be mathematically described by Eq. 27 [83]:

$$q_{ABL} = \left[\underbrace{(1 - \beta_R) \left(\frac{m}{2 \cdot \pi \cdot k_B \cdot T} \right)^{\frac{1}{2}} p_{rec}(T)}_{\text{evaporation rate } \dot{m}_v} \right] \left(\int C_p(T) dT + L_m + L_v \right) \quad (27)$$

where q_{ABL} the heat flux due to ablation in $[\text{W}/\text{m}^2]$, β_R a recombination rate, m represents the mass per atom of a metal in $[\text{kg}]$, L_v represents the latent heat of evaporation $[\text{J}/\text{kg}]$, L_m the latent heat of melting $[\text{J}/\text{kg}]$ and k_B is the Boltzmann constant. The pressure $p_{rec}(T)$ on the surface is calculated by the expression:

$$p_{rec}(T) = p_o \cdot \exp \left[\frac{m \cdot L_v}{k_B \cdot T_v} \left(1 - \frac{T_v}{T} \right) \right] \quad (28)$$

with p_o the ambient pressure, 1.01×10^5 $[\text{Pa}]$, and T_v material's boiling point in $[\text{K}]$.

2.3 Powder Bed Thermophysical Properties

In order to fully model the heat transfer of SLM process, the material's thermophysical properties have to be determined. As it is well known, thermophysical properties are temperature dependent, thus, they have to be specified according to the temperature. Since in this way additional non-linearity is introduced into the model, in sake of simplicity and to spare computational time, researchers have presented models using constant thermophysical properties

[84]. Nevertheless, this approach is best to be avoided. In a more realistic approach, thermophysical properties are estimated according to the powder bed porosity, taking in mind that powder consist of solid granules and a fluid material (gas). Here, different approaches have been proposed, some simpler and some more complex. For example, Foroozmehr et al. [37] considered the thermal conductivity of the powder bed equal to 1% of the bulk material. Unquestionably, the major is to adopt a modeling strategy that takes in mind the peculiarities of a powder medium regarding its thermophysical properties.

In the study of Hussein et al. [29] the porosity of the powder bed and the powder thermal conductivity are calculated as:

$$\varphi = \frac{\rho_s - \rho_{PB}}{\rho_s} \Rightarrow \rho_{PB} = (1 - \varphi) \rho_s, \quad k_{PB} = k_s (1 - \varphi) \quad (29)$$

with ρ_s and ρ_{PB} the density of the solid material and powder on $[\text{kg}/\text{m}^3]$ respectively, while k_s is the solid material thermal conductivity and k_{PB} the thermal conductivity of the powder. Antony et al. [28] assume that the powder bed consists of spherical, equal in size and densely arranged particles, estimating the powder's porosity and conductivity as:

$$k_{PB} = \frac{\pi k_s}{6 \left(1 + \Phi \frac{k_s}{k_g} \right)} \quad (30)$$

with Φ empirical coefficient $\Phi = 0.02 \cdot 10^{2 \left(0.7 - \frac{\pi}{6} \right)}$ and k_g the thermal conductivity of the surrounding gaseous environment in $[\text{J}/\text{kgK}]$. In a more detailed definition of the thermophysical properties, Loh et al. [23] not only took into consideration the different density and porosity of the powder bed, but the occurring phase change as well. In their work, a simplified relation about the powder bed thermal conductivity is implemented:

$$k_{PB} = k_s (1 - \varphi) \Phi = 4(\text{empirical coefficient}) \quad (31)$$

while the properties of material, during the phase changes, are calculated based on the volumetric ratio θ of each phase:

$$\begin{aligned} k_{\text{phase}1,2} &= \theta k_{\text{phase}1} + (1 - \theta) k_{\text{phase}2} \\ C_{p_{\text{phase}1,2}} &= \theta C_{p_{\text{phase}1}} + (1 - \theta) C_{p_{\text{phase}2}} + L_{\text{phase}1 \rightarrow 2} \frac{d\theta}{dT} \end{aligned} \quad (32)$$

with $L_{\text{phase}1 \rightarrow 2}$ the latent heat of phase change. At this point, it is important to emphasize on the consideration of latent heat of melting and evaporation, as phase changes consume considerable amount of energy. Especially on the boundaries of the melt pool, the calculation of latent heat, may significantly affect the results and conclusions about the melt pool geometry and dimensions. Thus, latent heats of melting and evaporation have always to be included in simulation.

Finally, one of the most thoroughgoing models about estimation of the powder bed's thermal conductivity is the Zehner—Schlunder model [81]. Based on the work of Sih and Barlow [82], the thermal conductivity of the powder bed can be calculated as:

$$k_{PB} = k_{ff} + k_{core} + k_{contact} \quad (33)$$

$$k_{ff} = k_g \left(1 - \sqrt{1 - \phi} \right) \left(1 + \frac{k_R}{k_g} \right) \quad (34)$$

$$k_{core} = k_g \sqrt{1 - \phi} \left\{ (1 - \phi) \left[\frac{2}{1 - \frac{B \cdot k_g}{k_s}} \left(\frac{B}{\left(1 - \frac{B \cdot k_g}{k_s} \right)^2} \left(1 - \frac{k_g}{k_s} \right) \ln \frac{k_g}{B \cdot k_s} - \frac{B + 1}{2} - \frac{B - 1}{1 - \frac{B \cdot k_g}{k_s}} \right) + \frac{k_R}{k_g} \right] \right\} \quad (35)$$

$$k_{contact} = \begin{cases} \phi^{\frac{18\phi k_s}{k_g}} \text{ for } \phi < 3 \cdot 10^{-4} \\ \phi^{\frac{k_s}{k_g}} \text{ for } \phi > 0.01 \end{cases} \quad (36)$$

where k_{ff} represents the free fluid radiation heat transfer, k_{core} the core heat transfer, $k_{contact}$ the solid contact heat transfer, k_R is the part of thermal conductivity owing to radiation, B is the deformation parameter of the particle according to its geometry, and $k_{sol.con}$ represents the solid contact conductivity. The deformation parameter B in the present case is expressed as follows:

$$B \approx 1.25 \left(\frac{1 - \phi}{\phi} \right)^{\frac{10}{9}} \quad (37)$$

The thermal conductivity part due to radiation is calculated as follows:

$$k_R = \frac{4 \cdot \varepsilon \cdot k_{Boltz} \cdot T^3 \cdot d_R}{1 - 0.132 \cdot \varepsilon_{T < T_m}} \quad (38)$$

Up to here, the main modeling techniques and strategies in simulation of heat transfer during SLM were presented. Of course, and as modeling of SLM is an open scientific topic, researchers are asked to present and propose improved variations of those models, with even more accurate and realistic results.

3 Melt Flow Dynamics

As researchers have access on an increasing computational power, multidisciplinary and multi-physics models are presented, coupling thermal analysis with fluid dynamics, in a more thorough approach. The melt pool flow is complicated

and numerous different mechanisms have to be taken into consideration, thus there is a margin for assumptions, simplifications and justified "modeling tricks" in order for the simulation to become feasible in a reasonable time. Nevertheless, there are some common principles in modeling of melt pool fluid dynamics during SLM that will be presented in the following section.

At first, the mass conservation is expressed as [78]:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) = M_s \quad (39)$$

with M_s the mass source and $\vec{V}$ the velocity vector. Since during SLM no mass source exists, $M_s = 0$ [34]. Moreover, assuming that the molten material is Newtonian and incompressible, Eq. 39 can be simplified as [31]:

$$\nabla \cdot \vec{V} = 0 \quad (40)$$

In the case that the model incorporates calculations about the material evaporation, two parameters, α_1 and α_2 , have to be employed to simulate the phase change. To fulfil the conservation of mass principle, the summation of α_1 and α_2 has to be always unity ($\alpha_1 + \alpha_2 = 1$). Additionally, a liquid–gas interface is resulted, which has to be defined. The evolution of the interface can be calculated using the evaporation rate as [85]:

$$\frac{\partial a_1}{\partial t} + \nabla \cdot (a_1 \vec{V}) = -\frac{\dot{m}_v}{\rho_v} \quad (41)$$

with $\dot{m}_v$ the mass evaporation rate in [kg/s] which can be calculated according to Eq. 27, and ρ_2 the density of metal vapor in [kg/m³].

The conservation of momentum (Navier–Stokes equation) can be expressed as [86]:

$$\frac{\partial \rho \vec{V}}{\partial t} + \nabla \cdot (\rho \vec{V} \otimes \vec{V}) = \mu \nabla^2 \vec{V} - \nabla p + M_s \vec{V} + \vec{F} \quad \text{with } \vec{F} = \vec{F}_{buoy} + \vec{F}_{Darcy} + \vec{F}_{surf} \quad (42)$$

with μ the viscosity in [kg/m·s] and $\vec{F}$ the vector of forces that act on the fluid volume. The forces include buoyancy forces caused by density differences, due to thermal expansion, Darcy's term as the damping term in the mushy zone and the surface forces. The buoyancy forces by employing the Boussinesq approximation can be expressed as [60]:

$$\vec{F}_{buoy} = \rho g \vec{e}_z \beta (T - T_{ref}) \quad (43)$$

with $\vec{e}_z$ the unit normal of gravitational force, β the thermal expansion coefficient, and T_{ref} the reference temperature. The Darcy's term is calculated based on the Carman – Kozeny equation [85, 87]:

$$\vec{F}_{Darcy} = -K_C \left[\frac{(1-f_l)^2}{f_l^3 + C_K} \right] \vec{V} \quad (44)$$

with K_C a constant referred to the permeability of the porous media, f_l the fraction of liquid metal (in the liquid phase $f_l=1$), and C_K a small numerical constant to avoid division by zero in case of $f_l=0$. Finally, the surface forces include a surface tension term, i.e. capillary force, the Marangoni force, i.e. thermo-capillary force, and the recoil pressure. The surface forces can be expressed according to Eq. 45 [60, 85] as:

$$\vec{F}_{surf} = \left\{ \underbrace{\sigma_{surf} \kappa \vec{n}}_{\text{capillary force}} + \underbrace{\frac{d\sigma_{surf}}{dT} (\vec{\nabla} T - \vec{n}(\vec{n} \cdot \vec{\nabla} T))}_{\text{thermo-capillary force}} + \underbrace{p_o \cdot \exp \left[\frac{m \cdot L_v}{k_B \cdot T_v} \left(1 - \frac{T_v}{T} \right) \right]}_{\text{recoil pressure (see eq. 28)}} \right\} \cdot |\vec{\nabla} a_1| \cdot \frac{2\rho}{\rho_s + \rho_g} \quad (45)$$

with σ_{surf} the surface tension acting on a surface curve κ , and $\vec{n}$ the unit normal. The term $|\vec{\nabla} a_1|$ indicates that all the sur-

face forces are acting on the liquid–gas interface, while the $\frac{2\rho}{\rho_s + \rho_g}$ term is used to smear out between hard and soft phases, with ρ_g the density of the gaseous phase. At this point, a remark has to be made regarding the pressure. The pressure is defined as the average normal stress, thus the trace of the deviatoric stress tensor is zero. On the contrary, if the pressure is defined as the thermodynamic pressure, then the trace of the deviatoric stress tensor is not zero, and has to be taken into consideration [85]. The convective intensity of Marangoni flow can be described by the Marangoni number (Ma), a dimensionless parameters, which is expressed as [78, 88]:

$$Ma = -\frac{\partial \sigma_{surf}}{\partial T} \cdot \frac{\Delta T \cdot L}{\mu D} \quad (46)$$

with D the thermal diffusivity in $[m^2/s]$, and L the character-

istic length (liquid layer thickness).

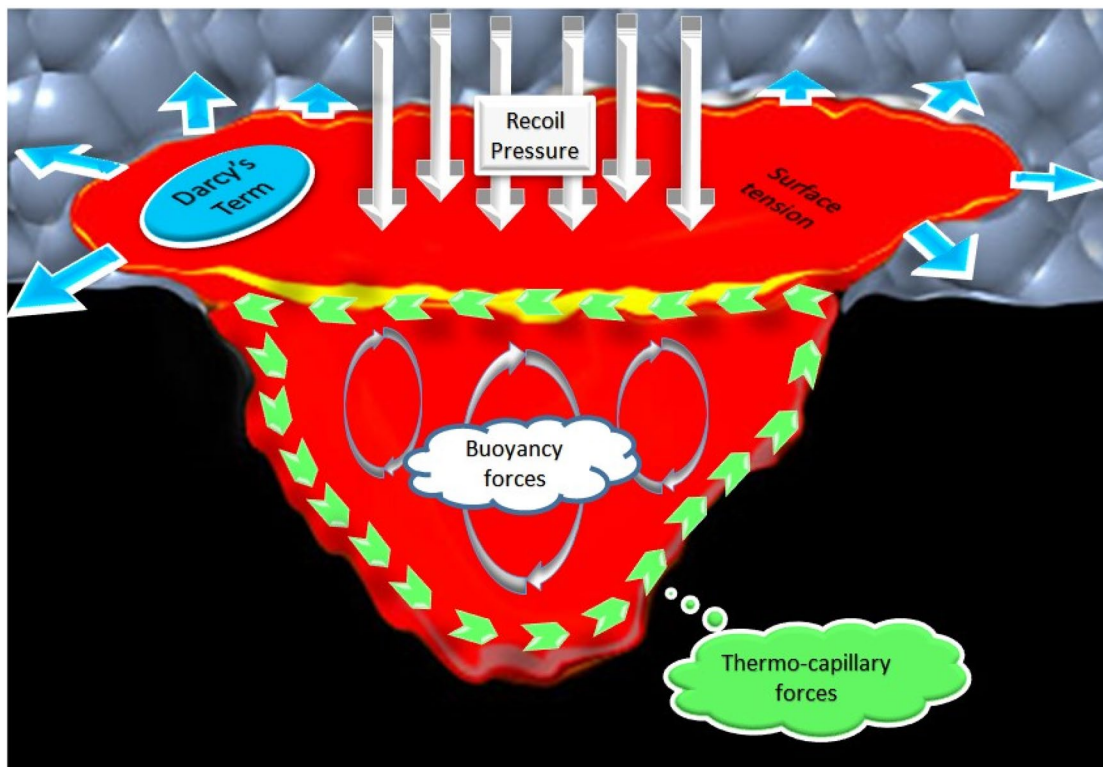


Fig. 3 Graphical presentation of the main forces that act on the melt pool

In Fig. 3 the main forces that act on the melted material volume are graphically represented.

Regarding the modeling of the interaction of various phases existing in the melt pool, special methods should be applied in addition to the fluid flow equations in order to track the interface between different phases. The most important methods employed for this reason are the Arbitrary Lagrangian Eulerian (ALE) method, the Level set method and the Volume of Fluid method. The ALE method combines the capability of modeling large deformations due to the Eulerian description which employed a fixed mesh and the capability of accurately representing the free surfaces and interfaces due to the Lagrangian description which employs a mesh that evolves according to the movement of the material. [89] After Lagrangian motion is performed, Eulerian steps are employed with mesh relaxation and state and material properties are remapped on the updated mesh. In the case of ALE models, element surfaces can cross material boundaries and thus mixed elements are created, something that does not occur in pure Lagrangian models.

The level set method is another method for the tracking of an interface within a continuum which was discretized using the FE method. The definition of the interface is performed using a special function ψ which is a function of both space and time and refers to the distance of any point at any time from the interface which is represented by the value $\psi=0$. In the transition zone between the two domains divided by the interface the physical properties are calculated by a mixture law. After the velocity field is calculated in the total domain during each step of the numerical solution, a differential equation is solved in order to reconstruct the interface [90, 91].

Finally, using the VOF method, the tracking of the free surface is conducted by solving a modified continuity equation for the volume fractions of different phases, given in Eq. (40). The variable related to the volume fraction in the cases of two different phases has values in the range [0,1] with 0 corresponding to the pure first phase, 1 to the pure second phase and intermediate values indicating the existence of both phases with an interface between them. In the latter case, the various quantities are calculated using a mixture law [92, 93].

In a brief note about fluid dynamics physical phenomena and their modeling, the Boussinesq approximation is referred on the assumption that all fluid properties other than the density are completely ignored [94]. Variations of the density are ignored except insofar as they give rise to a gravitational force. Hence, the continuity equation can be simplified in Eq. 39, and the buoyancy forces term of Eq. 43 is resulted. The thermo-capillary forces are part of Marangoni effects, which were initially explained by Thomson [95]. In Marangoni effects, the difference in interfacial tension, along the interface between two media, results to a

secondary flow, which is generated from a lower-interfacial tension area to a higher-interfacial tension area. Depending on the cause of the interfacial tension difference, Marangoni effects are characterized as thermo-capillary effect utilizing a temperature gradient, solutal-capillary effect if they are resulted by concentration difference and chromo-capillary effects in cases of photosensitive surfactants and UV light [96]. In SLM, only the thermo-capillary forces are taken into consideration, as high gradients of temperature occur during the process, giving rise in thermo-capillary convection. Finally, Darcy's law is a phenomenologically derived constitutive equation that describes the flow of a fluid through a porous medium. Darcy's law is valid for slow, viscous flow, like the molten material during SLM. Conceptually, Darcy's law at constant elevation is a simple proportional relationship between the instantaneous flow rate through a porous medium, the viscosity of the fluid, and the pressure drop over a characteristic distance [97]. During SLM, the molten material constitutes the fluid, while the powder bed represents the porous medium.

In the relevant literature, studies with only thermal modeling, without integrating fluid dynamics can be found. This approach is not wrong, serving the need for simpler simulations by reducing the demands in computational time and power. The main criterion for such a decision is the melting mode during SLM. In conduction melting mode, where relatively low velocities are developed within the melt pool, while heat conduction is the major mechanism, flow—related phenomena can be neglected, adopting a thermal model. In many studies, the melt pool dimensions are predicted, with high accuracy, based on a thermal model [69, 71, 72]. However, when flow—related phenomena, as convection flow, Marangoni convection, Plateau-Rayleigh capillary instability etc. are dominant, the implementation of a multi-physics model, by coupling heat transfer with fluid dynamics is necessary, on behalf of realistic and accurate results.

In brief, the keyhole mode is observed when the recoil pressure that acts on the surface of the molten pool exceeds the pressure produced by the surface tension, and thus, the molten material is pushed out of the cavity [84]. The dependence of the keyhole threshold from the processing parameters can be estimated in terms of normalized enthalpy. Specifically, the absorbed energy during a dwell time t_{dwell} , results to an average absorbed energy density of [84]:

$$\text{Energy Density} = \frac{\eta P t_{\text{dwell}}}{\pi r_{LB}^2 \sqrt{D t_{\text{dwell}}}} \quad \text{with} \quad t_{\text{dwell}} \sim \frac{r_{LB}}{u_{LB}} \quad (47)$$

with u_{LB} the laser's scanning speed.

Thus, the specific enthalpy ΔH divided by the enthalpy at the melting point h_{melt} can be expressed as [84]:

$$\frac{\Delta H}{h_s} = \frac{\eta P}{\pi h_s \sqrt{Du_{LB} r_{LB}^3}} \quad \text{with} \quad h_{melt} \approx \rho C_p T_m \quad (48)$$

A calibration of the normalized enthalpy is often employed [84, 98]:

$$\frac{\Delta H}{h_s} = \frac{\eta P}{h_s \sqrt{\pi Du_{LB} r_{LB}^3}} \quad (49)$$

Finally, the estimation criterion for the keyhole threshold can be obtained as [84]:

$$\frac{\Delta H}{h_{melt}} > \pi \frac{T_v}{T_m} \quad (50)$$

The heat transfer equations combined with those of fluid dynamics and mass transfer can adequately describe and simulate both, the conduction and the keyhole mode. Nevertheless, and as it is already mentioned, aiming to reduce the demands in computational power and time, their use must be case driven. In some particular cases, more straightforward

models can be adopted (e.g., simpler heat transfer models), while in some others there is an absolute necessity to employ the equations of heat transfer, along with those of fluid dynamics and mass transfer. After all, the ultimate criterion for every model is its capability to be in line with the respective experimental results.

4 Residual Stresses

As, by its nature, SLM is a rapid process, it is featured by high temperature gradients, along with extreme heating and cooling rates. Due to the contraction and extension of the material, residual thermal stresses are developed [99]. In Fig. 4 the main stress fields that are developed during SLM are depicted.

To model residual stresses, the governing stress equilibrium equations are required [26]:

$$\nabla \sigma = 0 \quad (51)$$

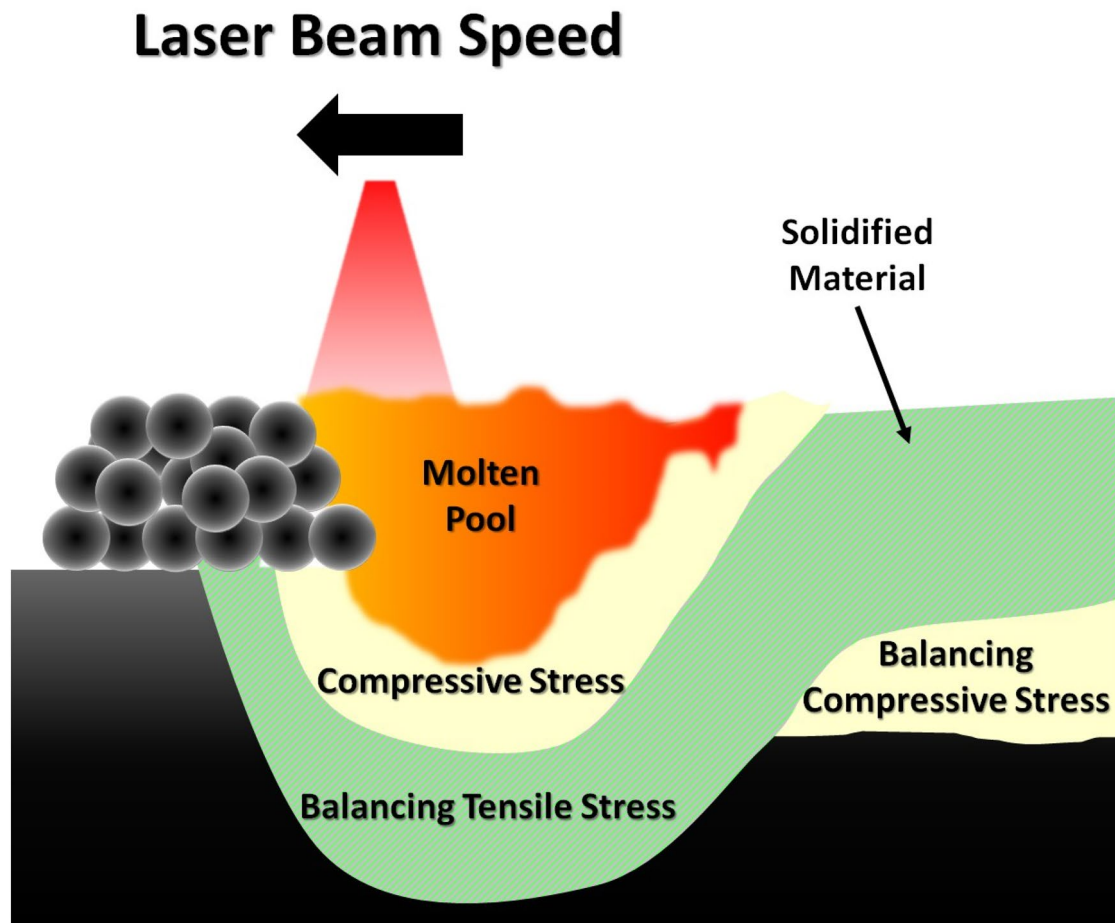


Fig. 4 Graphical presentation of the main developed stress fields in SLM

with σ the stress tensor. The constitutive equation that correlates stress, strain and material properties is [26]:

$$\{\sigma\} = [K]\{\epsilon^{el}\} \quad (52)$$

with $\{\sigma\}$ The stress vector, $[K]$ the elasticity matrix, and ϵ^{el} the elastic strain.

In some studies the assumption of isotropic material and axial symmetry have been made [26, 29, 100], and thus, Eq. 52 is simplified and can be written in Cartesian coordinates as:

$$\begin{aligned} \epsilon_{xx} &= \frac{1}{E} [\sigma_{xx} - \nu(\sigma_{yy} + \sigma_{zz})] + a_{th} \Delta T \\ \epsilon_{yy} &= \frac{1}{E} [\sigma_{yy} - \nu(\sigma_{xx} + \sigma_{zz})] + a_{th} \Delta T \\ \epsilon_{zz} &= \frac{1}{E} [\sigma_{zz} - \nu(\sigma_{xx} + \sigma_{yy})] + a_{th} \Delta T \\ \epsilon_{xy} &= \frac{1+\nu}{E} \sigma_{xy}, \epsilon_{xz} = \frac{1+\nu}{E} \sigma_{xz}, \epsilon_{yz} = \frac{1+\nu}{E} \sigma_{yz} \end{aligned} \quad (53)$$

with E the modulus of elasticity, ν the Poisson's ratio and a_{th} the thermal expansion coefficient in $[K^{-1}]$.

By implementing the small deformation theory the total strain can be calculated as [13]:

$$\epsilon^{tot} = \epsilon^{el} + \epsilon^{pl} + \epsilon^{th} \quad (54)$$

with ϵ^{tot} , ϵ^{pl} , and ϵ^{th} the total, plastic and thermal strain respectively. The thermal strain can be computed as [26]:

$$\epsilon^{th} = a_{th}(T - T_{ref}) \rightarrow a_{th} \text{ is temperature dependent } \epsilon^{th} = \int_{T_{ref}}^T a(T) dT \quad (55)$$

The yield of material occurs when $f^{yield} \leq 0$ with f^{yield} defined as [25]:

$$f^{yield} = \sigma_{VM} - \sigma_Y(T) \quad (56)$$

with σ_Y the temperature dependent material's yield strength, and σ_{VM} the von Mises stress, calculated as:

$$\sigma_{VM} = \sqrt{\frac{1}{2} [(\sigma_{xx} - \sigma_{yy})^2 + (\sigma_{yy} - \sigma_{zz})^2 + (\sigma_{zz} - \sigma_{xx})^2 + 3(\sigma_{xy}^2 + \sigma_{yz}^2 + \sigma_{zx}^2)]} \quad (57)$$

At this point some important details have to be pointed out. During SLM the material changes phases, passing from powder material to molten and then it is solidified. Mechanical stresses can be developed only during solid phase and more specifically, for temperatures above the stress relaxation temperature (T_{relax}). Thus, a relaxation temperature has to be defined, above which, strain components are set to zero [24]. This way, not only the annealing can be modeled, but also the inability of developing mechanical stresses during the molten phase. The second point that has to be emphasized is the material shrinkage during melting. Molten material sinks as pores that initially exist between the powder particles that are removed. The material's volume shrinkage

is analogue to the powder's porosity and occurs gradually from the powder surface to the depth of the powder bed [23, 101, 102]. Aiming in detailed and accurate analysis results, this volume shrinkage has to be taken into consideration in modeling.

5 Microstructure modelling

Additionally, modeling of SLM provides the possibility to model the microstructure of the processed material. There are three microstructure models that are mainly employed: Phase Field modeling (PFM), kinetic Monte Carlo (MC), and Cellurral Automata (CA). MC and CA are meso-scale modeling methods, simulating multiple grains, considering the overall size and aspects ratio of the formed grains. On the contrary, PFM is a micro-scale modeling method, simulating phenomena that occur in sub-grain level, i.e. the solute concentration, precipitates and dendrites shape. Moreover, multiple grains can be simulated by using multi-phase field models.

In PFM, the main principle is that a fixed volume system during solidification would achieve stability by lowering its free energy. Mathematically it can be expressed as [103]:

$$E^{free} = \int [f_{th,d}(\phi_{Ph,F}, c_{Ph}, T) + \omega g(\phi_{Ph,F}) + \epsilon_{Ph,h}^2 |\nabla \phi_{Ph,h}^2|] dV_{ol} \quad (58)$$

with E^{free} the total system's free energy, $f_{Ph,F}$ the thermodynamic potential, $\phi_{Ph,F} \in [-1, 1]$ the phase field variable, $c_{Ph,F}$ the concentration, $\omega g(\phi_{Ph,F})$ the double well potential, and $\epsilon_{Ph,h}^2 |\nabla \phi_{Ph,h}^2|$ the phase field energy gradient in V_{ol} volume. The phase-field gradient energy depends on the gradient of the interface, thus would be zero in fully solid and liquid regions. The thermodynamic potential is defined as the sum of the free energies of the liquid and solid states. In time

dependent systems, Allen – Chan (Eq. 57) and Cahn Hilliard (Eq. 58) equations “guarantee” the decrease in total free energy with time [104]:

$$\frac{\partial \phi_{Ph,F}}{\partial t} = -M_{\phi_{Ph,F}} \left[\frac{\partial f_{th,d}}{\partial \phi_{Ph,F}} + \omega g'(\phi_{Ph,F}) - \epsilon_{Ph,h}^2 \nabla \phi_{Ph,h}^2 \right] \quad (59)$$

$$\frac{\partial c_{Ph}}{\partial t} = \nabla \cdot M_{c_{Ph}} \nabla \frac{\partial f_{th,d}}{\partial c_{Ph}} \quad (60)$$

with $M_{\phi_{Ph,F}}$ and $M_{c_{Ph}}$ the motilities of the phase field and concentration field at the interface respectively. In a multiphase system the sum of phases has to reach unity. Moreover, all

variables have to be continuous across the interface, while the interface can be a fraction of both solid and liquid.

Conceptually, MC is an on-lattice modeling method to simulate the curvature driven grain growth, based on the minimization of the system's energy. A polycrystalline microstructure is approximated by assigning an integer "spin" to each lattice site. Grains are composed by continuous sites of the same spin. Neighbors with unlike spins contribute to the system's total energy, while there is no change in the system energy if neighbors have the same spin. Thus, the local configurational energy is determined as [105]:

$$E^{\text{sys}t} = \frac{1}{2} \sum_{i=1}^{\text{number of lattice sites}} \sum_{j=1}^{\text{number of neighbors}} [1 - \delta(\text{spin}_i, \text{spin}_j)] \quad (61)$$

with δ the Kronecker delta function, namely, δ is equal to 1 when $\text{spin}_i = \text{spin}_j$ and 0 otherwise. The number of neighbors is equal to 26 or 8 in case of 3D or 2D domains, respectively. System's energy minimization can be obtained through curvature-driven grain growth. Each site has a probability of changing its spin. If the spin change contributes to a decrease in the overall system's energy, then it is accepted. On the other hand, if the spin change causes an increase in energy, a random number would be generated for that site, and will be compared to an acceptance probability. Only if the random number is smaller than the acceptance probability, the new spin will be accepted. The acceptance probability is calculated as [105]:

$$\text{Prop.} = \begin{cases} \exp\left(-\frac{\Delta E^{\text{sys}t}}{k_B T}\right) & \text{if } \Delta E^{\text{sys}t} > 0 \\ 1 & \text{if } \Delta E^{\text{sys}t} \leq 0 \end{cases} \quad (62)$$

In solidification modeling, Eq. 62 is modified, and the acceptance probability Prop. depends on the temperature, following the Arrhenius equation [103]:

$$\text{Prop.} = \begin{cases} \left[M_0 \exp\left(-\frac{Q_{\text{act}}}{R_{\text{gas}} T}\right) \right] \exp\left(-\frac{\Delta E^{\text{sys}t}}{k_B T_{\infty}}\right) & \text{if } \Delta E^{\text{sys}t} > 0 \\ M_0 \exp\left(-\frac{Q_{\text{act}}}{R_{\text{gas}} T}\right) & \text{if } \Delta E^{\text{sys}t} \leq 0 \end{cases} \quad (63)$$

with M_0 the Arrhenius pre-exponential factor, Q_{act} the activation energy and R_{gas} , the gas constant.

Finally, the CA is based on equally divided cells, i.e., square in 2D models and cube for 3D, with each cell being interactive with its neighboring ones. In the case of solidification, the material's state changes between solid, liquid and mushy. In CA, the governing equations regarding the transition from liquid to solid and vice versa have to be defined, to calculate the crystallographic orientation, as well the speed and orientation of capturing neighboring cells for crystal growth. CA combines the rules of solidification with

some probability elements for nucleation, since the change from liquid to solid state does not depend only on temperature, but also on the probability of nucleation as well. According to CA probabilistic approach, the grain density η^{grain} for an undercooling ΔT that drives the solidification is calculated as [103]:

$$\eta^{\text{grain}}(\Delta T) = \int_0^{\Delta T} \frac{d\eta^{\text{grain}}}{d(\Delta T)} dT \quad (64)$$

while a Gaussian distribution is employed to determine if a cell will form a nucleus [103]:

$$\frac{d\eta^{\text{grain}}}{d(\Delta T)} = \frac{\eta_{\text{max}}^{\text{grain}}}{\sqrt{2\pi} \exp\left[-\frac{(\Delta T - \Delta T_{\text{mean}})^2}{2\Delta T_{\sigma}^2}\right]} \quad (65)$$

with $\eta_{\text{max}}^{\text{grain}}$ the maximum nucleation density, ΔT_{mean} the mean undercooling when the nucleation occurs, and ΔT_{σ} the undercooling standard deviation. The grain growth is determined by solidification theory, hence, a growth kinetic model, e.g. Kurz-Giovanola-Trivedi (KGT) model, has to be adopted and integrated. Often, CA is coupled with macro-thermal FEM. Similar with the MC models, dendritic structures like secondary arm growth are not calculated, but instead, the grain structure is simulated allowing computations in a larger spatial scale.

6 Modeling and Simulation Obtained Results

Modeling and simulation of SLM are powerful tools in a depth analysis and understanding of the process, as they give an inmost view of the underlying mechanisms, which occur rapidly, thus, the in situ experimental observation is difficult or even impossible. The obtained results and information are depending on the implemented modeling strategies and assumptions, as they are derived straight from solving model's governing equations and its boundary conditions. Therefore, results regarding the developing temperatures, melt pool dynamics and residual stresses are expected.

A clarification concerning the extent of simulation is important and necessary; in the overwhelming majority of modeling studies single tracks (one or several) are simulated and not the entire building process. In reality, the manufacturing of a component demands hundreds or even thousands of subsequent tracks, for which modeling and simulation would require immense computational power and time. To become the simulation feasible, only a number of tracks are modeled, assuming that the process is repetitive, thus, the obtained results have a general implementation. By

simplifying the process environment, the control volume is discrete into the substrate and the powder bed [106], while, in a more precise description, the shielding gas is also taken into consideration [73]. According to the selected modeling strategy, material properties are properly defined, namely, the substrate owns bulk material properties, whilst the powder bed incorporates the powder phase properties that have been earlier discussed. Between the different phases and materials, the continuity and conservation of energy and mass have to be assured. In the following section, the most important results of modeling and simulation will be presented and summarized.

The fundamental result is the determination of the temperature profile, spatially and temporally. According to spatial distribution, temperature gradients can be easily calculated, and subsequently, a comparison with experimental results can be made [50, 106]. From the temporal temperature distribution, heating and cooling rates can be estimated; these are essential information about the process, since they define the material's microstructural characteristics, and thus its acquired mechanical properties [39, 99]. Based on the temperature profile, the geometrical characteristics of the melt pool can be determined, which at the same time are employed as a common and reliable simulation's validation method. Most often, the width, depth and length of the melt pool are presented [36], while in some studies, the molten material volume is calculated as well [75]. The definition of the melt pool dimension is based on the temperature profile and the premise that material's temperature has exceeded its melting point. An important detail has to be pointed out regarding the melt pool depth; it is important to pay attention on how the melt pool depth is defined and measured. In some studies, the top plain of the powder bed is considered as the reference plain [37, 86, 107], whereas, in other studies, the substrate top plain is taken as reference, while the overhead material is defined as track's height [39, 57, 108]. The significance of that detail lays on the avoidance of any over/under estimation of melt pool dimensions. The accurate calculation of the melt pool dimensions is of extreme interest, since melt pool's stability greatly depends on its geometrical characteristics. More specifically, in the work of Zhang et al. [36, 43] the Plateau-Rayleigh instability is predicted using the quotient of the melt pool circumference to its length. Moreover, melt pool's ripple angle can be estimated [36], as well the isothermal contours [58] and peak temperatures during SLM [109]. The melt pool geometrical characteristics are figured out for different machining parameters, i.e. laser beam power, scanning speed, powder bed thickness, different materials, and preheating temperatures. Furthermore, in studies where multiple subsequent tracks are simulated, the hatch space is an additional manufacturing parameter, while the overlapping between tracks can be estimated as well [37, 57, 110].

Simulations that integrate fluid dynamics can obtain more data and information about the process. Firstly, the velocity fields can be depicted, representing the molten material flux during SLM [111], while the recoil pressure along the track can be calculated, too. Additionally to these primary results, exceptionally useful secondary conclusions can be drawn, giving an inner view of the process, and allowing a more in depth analysis. Often, and as the available computational power increases, the simulation is combined with implementation of dynamic mesh methods, i.e. ALE, Lagrangian mesh, deformed geometry feature, meaning the capability of tracking surface geometrical changes during SLM. Thus, an augmented supervision of the process is feasible, including the study of phase change and vaporization phenomena. Melt pool front evolution is of the most interest, therefore, many studies are focusing on presenting the occurring mechanisms. Specifically, the velocity field and the evaporated material result to morphological changes in the surface, changes that can be depicted by modeling the melt pool dynamics [73, 112]. Solid-liquid interfaces can be estimated in different stages of the melt pool evolution [113], as well its discrete regions of depression, transition and tail—end. In these regions, a non-uniform complex flux is developed, resulting to characteristic features of SLM, like the keyhole effect, denudation areas, balling effects and pore forming [114]. As modeling becomes more detailed and comprehensive, phenomena like material expulsion, laser's shadowing due to spattering, multiple keyholes resulted by the scan strategy and defects formation can be simulated and studied, while computational results are in line with experimental ones [115, 116]. Additional to these main results, secondary conclusions can be drawn regarding the process. The vaporized material flow (plume) can be simulated [62], allowing an estimation of the topical changes in material properties and calculations about the interaction of the laser beam with the formed plume, best known as plasma shielding mechanism [117]. Moreover, concurrently with the vaporized material flow, heat is getting away, thus, new energy equilibrium is taking place. Therefore, aiming on a detailed and accurate simulation, these parameters have to be taken into consideration, as they offer a thorough view of the process, enabling process' deeper understanding and its subsequent optimization.

Taking advantage of the spatial and temporal temperature field calculation, the developed strains and residual stresses can be estimated. Since residual stresses, strain and distortion are mainly related to the manufactured part, rather than a single track, models that integrate thermomechanical analysis are usually simulating multiple tracks [37], even deposited layers [25, 46, 72]. Substantial advantage of the thermomechanical modeling is the capability to compute the stress and strain fields for different scanning strategies [33], by changing the heat source path. Therefore, since the model

has been validated, different patterns and scanning strategies can be tested focusing on the process optimization. Finally, additional information can be obtained by simulating the SLM process. By integrating microstructure models and employing the calculated temperature history, a "grain map" can be obtained. The grain morphological characteristics, their density, orientation and temporal growth are estimated by coupling microstructural model with thermal or thermo-fluid one [13, 39, 60, 105, 107, 118]. The necessary models' validation and verification is performed by using techniques like SEM microscopy and EBSD.

Finally, a major challenge in the simulation of SLM is the coupling of different scales of modeling. The analysis can be ordered in micro-scale, meso-scale and macro-scale, where different spatial and time computational scales are studied [119]. In micro-scale, commonly, a single track is simulated, emphasizing on the melt flow characteristics, the geometrical and morphological characteristics of the track, as well on temperature and pressure fields that are developed. Going from micro- to meso- scale analysis, the studies are mainly focusing on multiple hatches, considering the temperature history and obtaining results regarding the porosity, the balling effect and the material's microstructure. In macro-scale modeling, usually, a thermal–mechanical analysis of the entire part is taking place, where residual stresses and part's distortions can be calculated [120, 121]. The major advantage of multi-scale modeling is that it offers the capability to have an insight of the process, from granule level up to the final 3D printed part.

7 Selective Laser Sintering

Regarding the modeling and simulation of SLS process, although it has many similarities with SLM modeling, there are some important differences that have to be accentuated. Both methods, i.e. SLM and SLS, are based on the utilization of a heat source (laser beam) of high power density to solidify a material, which is in powder state. The major difference is that in SLM the solidification presupposes the material melt, while in SLS it is achieved through sintering. Moreover, in SLM are primarily used metals, while in SLS polymers can also be used. The following section will be mainly focusing on the differences between modeling SLM and SLS, in order to avoid unnecessary repetitions.

Like in SLM, the SLS process simulation is based on a heat transfer model. The laser beam is approached by a heat source, however, in SLS, simpler models are usually adopted. A 2D Gaussian heat source, see Eq. 2, can be employed [122–124], or alternatively a 3D volumetric one. In volumetric heat source case, the power distribution along the powder bed thickness can be assumed as uniform

[125], or can be estimated according to Beer Lambert's law [126, 127]:

$$Q_{LB}(x, y, z) = \frac{2P\eta}{\pi r_{LB}^2} \cdot e^{-2\left(\frac{x^2+y^2}{r_{LB}^2}\right)} \eta^{BL} e^{-\eta^{BL}z} \quad (66)$$

with η^{BL} the extinction coefficient in $[m^{-1}]$ and the term $e^{-\eta^{BL}z}$ represents the exponential decay along the powder bed thickness. The material's absorption coefficient (η) is estimated experimentally, or semi-empirically through reverse engineering. Metallic and polymer materials have completely different optical properties, thus, since in SLS both can be processed, it is extremely important to accurately determine absorption coefficient, in respect to the laser beam wavelength. For example the reflectivity of CO₂ laser in polycarbonate power is approximately 4% [128], while the reflectivity in Ti is 97% [129]. The boundary conditions, namely the heat losses due to convection and radiation are estimated as in SLM, see Eqs. 18 and 23, employing the proper coefficients according to the materials and the process parameters, e.g. type of shielding gas [130].

An important parameter in SLS modeling is also the realistic estimation of the powder bed's thermal conductivity. In literature, many experimentally driven relations are applied, calculating the thermal conductivity according to material's porosity [131, 132]. The disadvantage of this approach is the need to define semi-empirical coefficients [123, 133]. As a rule of thumb, thermal conductivity is calculated based on the thermophysical properties of the solid and gas materials that form the powder bed, the porosity, and the sintering percentage. The equations are in a similar form with those that are employed in SLM, see Eqs. 29–37, after the necessary modifications according to the materials and their unique properties.

Another important aspect in SLS modeling is the latent heat. The Material is changing phase, from solid to liquid and vice versa, thus the latent heat of melting is an important parameter. The latent heat can be incorporated into material's specific heat [125, 134], thus, the specific heat is expressed as [135]:

$$C_p = C_{p_o} + \frac{L_{phase1 \rightarrow 2}}{\sqrt{\pi(T_{mf} - T_{ms})}} \exp\left(-\frac{(T - T_{ms})^2}{(T_{mf} - T_{ms})^2}\right) \quad (67)$$

or, in a small variation [136]:

$$C_p = C_{p_o} + \frac{L_{phase1 \rightarrow 2}}{\sqrt{\frac{\pi(T_{mf} - T_{ms})^2}{2}}} \exp\left(-\frac{2(T - T_{mp})^2}{\left(\frac{T_{mf} - T_{ms}}{2}\right)^2}\right) \quad (68)$$

with T_{ms} and T_{mf} the temperatures at onset and offset of melting respectively, T_{mp} the temperature at the peak point of

the heat capacity, and C_{Po} the heat capacity at the onset of melting.

By following another modeling method, the latent heat of melting, and the released temperature during the cooling stage due to crystallization (in case of polymer materials) are approached as a heat sink and a heat source respectively, contributing to the right term of the heat transfer equation, see Eq. 1. The heat sink can be mathematically expressed as [137]:

$$\dot{Q}_{\text{sink}} = -(1 - \phi)\delta h \frac{\partial f_{\text{liq}}}{\partial t} \quad (69)$$

with f_{liq} the volumetric fraction of liquid phase, and δh the total absorbed by the material latent heat. The volumetric fraction f_{liq} can be estimated in respect of the melting temperature range as [137]:

$$f_{\text{liq}} = \frac{(T - T_{ms})}{(T_{mf} - T_{ms})} \quad (70)$$

The heat source term due to crystallization can be modelled as [137]:

$$\dot{Q}_{\text{source}} = \rho_{\text{sol}} \frac{\partial a_{\text{cryst}}}{\partial t} \Delta H_{\text{cryst}} \quad (71)$$

with ρ_{sol} the solid material's density, ΔH_{cryst} the crystallization enthalpy and a_{cryst} the crystallinity rate. The crystallization degree $\alpha(t, T)$ as a function of time and temperature can be obtained by Nakamura's model [138, 139]:

$$\frac{\partial a_{\text{cryst}}}{\partial t} = \eta_{\text{cryst}} K_{\text{cryst}}(T) (1 - a_{\text{cryst}}) \left[\ln \left(\frac{1}{1 - a_{\text{cryst}}} \right) \right]^{\frac{\eta_{\text{cryst}} - 1}{\eta_{\text{cryst}}}} \quad (72)$$

with η_{cryst} the Avrami index, depends on the growth geometry of crystallites, and $K_{\text{cryst}}(T)$ the non-isothermal crystallization rate, which based on the Hoffman – Loritzen's theory is given by [137]:

$$K_{\text{cryst}}(T) = \ln(2)^{\frac{1}{\eta_{\text{cryst}}}} K_o \exp \left(-\frac{U}{R(T - T_G + 30)} \right) \exp \left(-\frac{K_G(T + T_{\text{fus}})}{2T^2(T_{\text{fus}} - T)} \right) \quad (73)$$

with U the activation energy of the crystallization transport taking the universal value of 6270 [J/mol], T_G the glass transition temperature, T_{fus} the equilibrium melting point in [K], and K_o , K_G , and η_{cryst} constants that are determined experimentally, by performing Differential Scanning Calorimetry (DSC).

Drummer et al. proposed a little different model to estimate the crystallization rate. Similarly with Nakamura's model, it needs DSC in order to be calibrated, while it is mathematically expressed as [140]:

$$\frac{\partial a_{\text{cryst}}}{\partial t} = \underbrace{a_{\text{cryst}}^{m_1} (1 - a_{\text{cryst}})^{n_1}}_{\text{Sestak - Berggren kinetic model}} \underbrace{Z_{\text{cryst}} \exp \left(-\frac{U}{RT} \right)}_{\text{Arrhenius equation}} \quad (74)$$

with m_1 and n_1 partial reaction order term and Z_{cryst} a pre-exponential factor in [min^{-1}].

Since crystallization concerns only the liquid material phase, the thermophysical properties of the fused part evolve during the cooling and can be estimated by following the mixing law, e.g. see Eq. 32.

A major difference between modeling SLS and SLM is that since in SLS the dominant solidification mechanism is sintering and not melting, fluid mechanics sub-models are rarely integrated into simulations. On the other hand, it is of extreme interest to calculate the sintering rate. Thus, relations concerning the sintering rate have been proposed and used. In the most commonly employed formula, the sintering rate is correlated with the material's density according to [128]:

$$\frac{\partial \rho}{\partial t} = (\rho_{\text{max}} - \rho) A_{\text{sint}} \exp \left(-\frac{U_{\text{sint}}}{RT} \right) \quad (75)$$

with ρ_{max} the density of a fully sintered material after infinite time, A_{sint} a pre-exponential factor and U_{sint} the sintering activation energy. This basic equation is modified, in respect of the material's unique properties. As an example, Childs et al. [131], in order to model the sintering of crystalline polymers, and the influence of the solid (crystalline) material fraction, proposed the calculation of the sintering rate according to:

$$\frac{\partial \rho}{\partial t} = (\rho_{\text{max}} - \rho) A_{\text{sint}} \exp \left(-\frac{U_{\text{sint}}}{RT} \right) \underbrace{\exp(-\beta_{\text{cryst}} f_{\text{sol}}^{n_2})}_{\text{retardation as the crystalline fraction increases}} \quad (76)$$

with β_{cryst} , and n_2 empirical coefficients, while f_{sol} the solid material fraction. Dong et al. [123] employed the same rela-

tion to estimate the sintering rate of Titanium, taking $n_2 = 1$.

Finally the shrinkage of the powder bed, according to mass conservation is calculated as [134]:

$$\text{shr} = \int_0^{z_{PB}} \frac{\rho_{PB}}{\rho(z)} dz \quad (77)$$

with shr the shrinkage ratio of a single layer with z_{PB} thickness. The shrinkage, along with the equations for stress

and strain calculation, see Eqs. 49–50, can estimate the developed stress and strain fields, as well any occurred deformation.

Like in SLM simulation, the obtained results are calculated according to the implemented governing equations. Since any SLS model is based on a heat transfer simulation, the temperature field, temperature gradients and the heating and cooling rates can be calculated [126]. Additionally, two crucial parameters, the maximum reached temperature during the process and the temperature history can be estimated [128]. Based on the spatial and temporal temperature field, the sintering depth and width can be defined in respect of the laser power and scanning speed [127], the hatching space if multiple tracks are simulated [136, 141, 142] and the laser frequency in case of pulsed laser [132]. Furthermore, by incorporating proper models, the density after sintering, i.e. the porosity and the degree of sintering are straight related with the density, as well as the crystallization rate can be calculated [137]. Finally, by coupling a heat transfer model with a thermomechanical one, the stress and strain fields, along with residual stresses and the occurred deformation, i.e. shrinkage, curling, warpage, can be defined [122, 139].

8 Modeling of Powder in SLM/SLS Processes

8.1 Discrete Element Modeling of Powder

Apart from the details regarding the modeling of thermal and fluid flow phenomena, residual stresses and microstructure alterations, there are also some other important issues, which should be taken into account for an accurate and realistic simulation of SLM and SLS processes. These issues are relevant to the modeling of the powder, which constitutes the raw material for SLM and SLS. More specifically, in simulations of both SLM and SLS, modeling of the powder layers, as well as the procedure of spreading the powder before it is irradiated by the laser beam, is essential to be conducted in order to achieve higher accuracy of the results and higher degree of realism of the simulation.

Thus, apart from the modeling of powder properties and behavior, discussed in previous subsections, the treatment of powder prior to its processing through SLM or SLS can be also modeled by appropriate numerical methods. In contrast to several models presented earlier, which consider the powder layer as a continuum domain although with suitable material properties, in the earliest stages of the additive manufacturing process, the discrete nature of the powder should be taken into consideration. In the later stages of the simulation, the powder domain can be also discretized with finite elements in the same way as the substrate and

solved according to the numerical schemes adopted for the FE analysis of the whole computational domain.

However, modeling of powder during its initial deposition and spreading cannot be performed by the FE method due to its discrete nature. Thus, a meshless method, which can account for the interactions between particles of the powder is required to be employed, with the most suitable being the Discrete Element Method (DEM). DEM was initially proposed by Cundall and Strack in 1979 in order to be applied to granular matter such as soil [143]. Since then, it has been employed in various simulations regarding granular matter in many engineering applications. DEM adopts a Lagrangian approach for the modeling and treats the granular medium as a number of distinct particles usually with a spherical shape with the same or different radii, moving according to the Newton's second law under the influence of forces of other particles or walls [144]. Interactions between non-spherical particles can also be modeled and they can be simplified appropriately into interactions of spherical particles [145]. Moreover, both translational and rotational motion of the powder particles can be taken into consideration [146]. Particles interact between them with forces occurring from contact or collision but also heat transfer and forming and breaking of bonds between particles can be modeled with DEM [147]. Thus, a DEM code should incorporate collision detection algorithms and particle contact models as well as means to handle boundary conditions and bonding between the particles [147]. In most cases, the DEM method is used for the accurate setup of the powder bed including the initial packing and the subsequent spreading process and afterwards the particles are integrated in the continuum simulation and discretized. Only in rare cases, the interactions due to thermal or other phenomena such as plasticity, phase change or shield gas flow field effect on particles movement are also taken into account [147–150]. In the present subsections, most of these details regarding modeling of powder particles by DEM method are presented.

More specifically, when powder particles are modeled by DEM method, the simplest approach for their modeling is to consider the effect of normal and tangential contact forces that are exerted on them. Then, the net force on each particle, due to their contact, is calculated according to the contribution of both forces. However, in many cases, more advanced models are used, taking into consideration additional force terms and multiphysics interactions as well [151]. In the present subsection, it is attempted to present most of the possible interaction types between powder particles that can be included in SLM/SLS simulations.

At first, the contact forces between particles can be calculated based on Hertz-Mindlin contact theory, whereas the cohesion force between particles can be calculated based on Johnson-Kendall-Roberts (JKR) cohesion theory [151]:

$$F_{ni} = (\kappa_n \delta_n^{3/2} + c_n \dot{\delta}_n) \frac{\delta_n}{|\delta_n|} \quad (78)$$

$$F_{ti} = (\kappa_t \delta_t + c_t \dot{\delta}_t, \mu_t F_n)_{\min} \frac{\delta_t}{|\delta_t|} \quad (79)$$

$$\mathbf{T} = \mathbf{r}_1 \times \mathbf{F}_t \quad (80)$$

$$F_{JKR} = -4 \left(\frac{\pi \alpha_{c.r.}^3 E \lambda_{k.e.}}{2(1 - \xi^2)} \right)^{0.5} \frac{\delta_n}{|\delta_n|} \quad (81)$$

where δ_n and δ_t represent the relative displacement in the normal and tangential direction, respectively, at the contact point between particles, μ_t is the friction coefficient for the sliding friction, $\alpha_{c.r.}$ is the contact radius and $\lambda_{k.e.}$ the surface kinetic energy. From the expression regarding the tangential force F_{ti} , it can be seen that it is bounded by the value of the Coulomb friction. The cohesion force actually represents the contribution of Van der Waals forces which are important for powders with size less than 100 μm and leads to cavity defects [146, 147, 152]. This force, as can be seen by the indices of some terms, is added to the normal contact force derived from the Hertz-Mindlin contact model [146]. Depending on the value of relative displacement or overlap in the normal direction, if it exceeds a critical value, it is suggested that the cohesive force contribution can be assumed as zero, in order to reduce computational cost [153, 154]. When the distance between the particles is lower than the critical value, the maximum value of cohesion force is attained, which is approximated as follows [153]:

$$F_{pullout} = \frac{3}{2} \pi R^* \gamma \quad (82)$$

where γ is the surface energy density. This force is termed as pull-out force and the comparison between this force and the gravity force defines the bonding number K . If the values of K are large, the particles exhibit high attraction forces [153, 154]. Alternatively, an expression based on Lennard-Jones potential function can be used to model the adhesive force [155, 156]. The coefficients κ_n , c_n , κ_t , c_t , representing the effective stiffness and damping coefficients in the normal and tangential directions can be further related to powder material properties as follows:

$$\kappa_n = \left(\frac{r_1 r_2}{r_1 + r_2} \right)^{0.5} \frac{2E}{3(1 - \xi^2)} \quad (83)$$

$$c_n = \left(\frac{m_1 m_2}{m_1 + m_2} \right)^{0.5} \frac{\ln e}{(\ln^2 e + \pi^2)^{0.5}} \left(\frac{5\kappa_n}{2} \right)^{0.5} \quad (84)$$

$$\kappa_t = \left(\frac{r_1 r_2}{r_1 + r_2} \right)^{0.5} \frac{2E}{(2 + \xi)(1 - \xi)} \quad (85)$$

$$c_t = \left(\frac{m_1 m_2}{m_1 + m_2} \right)^{0.5} \frac{\ln e}{(\ln^2 e + \pi^2)^{0.5}} \left(\frac{5\kappa_t}{3} \right)^{0.5} \quad (86)$$

where ξ represents the Poisson's ratio, e the restitution coefficient and E the Young's modulus of the powder particles. In the equations regarding κ_n and κ_t , the quantity $\frac{r_1 r_2}{r_1 + r_2}$ is also denoted as R^* which represents the effective radius. The terms including the stiffness and damping coefficients correspond actually to a spring-dashpot system for each particle pair.

The torque $\mathbf{T}$ presented in Eq. 80 is a result of the action of the tangential component of the contact force between two particles, which leads to their rotation. However, Gu et al. [146] considered also the rolling friction torque, which can restrain the rotation of a powder particle, as follows:

$$T_{ij}^r = -\mu_r R_i |F_{ij}^n| \omega_i \quad (87)$$

where μ_r represents the rolling friction coefficient and ω_i is a unit vector [146].

Apart from the modeling of the contact and cohesive forces between the particles, modeling of powder by DEM method includes various other elements. One important issue regarding the DEM simulation is the detection of collisions between particles [147]. Due to the relatively high number of particles required to model a powder bed, an efficient algorithm is required to detect the collisions. Many algorithms have been proposed for this reason but one of the most efficient ones is the approach using Verlet list and an octree spatial partitioning scheme [147]. With this method, for each particle, the list of particles, which are or may be in contact in the next steps within a specific radius, is determined after a fixed number of time steps. Then, forces due to collision of each particle with their surrounding particles are summed.

Another important feature of a DEM code is the determination of bond formation between the particles. The creation of a bond depends on parameters such as temperature, contact pressure and relative velocity of the particles [147]. At first, normal and shear forces, temperature and velocities of two particles, which are tested, are determined and then comparison between these values and critical values is performed to conclude whether the two particles are bonded or not. Similarly, when a pair of particles does not meet the requirements for bond formation any more, the bond between them is deleted. When the temperature is below the sintering temperature, every bond on a particle exerts a force on it [148]:

$$f_{B_{ij}} = k_B(\mathbf{p}_i - \mathbf{p}_{jL^2} - \delta_0) \quad (88)$$

where k_B is the bond stiffness and δ_0 is the bond length when the bond is unstretched. The bond is deleted when it reaches the failure strain [148]:

$$\left| \frac{\mathbf{p}_i - \mathbf{p}_{jL^2} - \delta_0}{\delta_0} \right| > \epsilon_\infty \quad (89)$$

Finally, the bond forces are summed to obtain the total force due to bonds, f_b .

When temperature is high enough and exceeds the melting point, the aforementioned constitutive law can no longer describe the behavior of particles, which will exhibit softening at higher temperatures. Then, it is more appropriate to use a material law such as the Johnson–Cook law for the particle bond forces. From the original formulation of Johnson–Cook law, the strain rate dependence can be neglected and finally, the bond force between two particles can be determined as follows [148]:

$$f_{B_{ij}} = a_{ij} \left(A + B \left(\frac{\mathbf{p}_i - \mathbf{p}_j - \delta_0}{\delta_0} \right) \right) \left(1 - \left(\frac{T_{ij} - T_0}{T_m - T_0} \right)^m \right) \quad (90)$$

where A , B and m are constitutive properties of the particle material, $a_{ij} = \pi * \left(\frac{r_i + r_j}{2} \right)^2$ represents the bond cross-sectional area, T_0 is the reference temperature and T_m is the material melting temperature.

When the simulation includes the effect of temperature on the particles, the solid part that occurs from the melting, bonding and solidification process should be predicted using the model. In order to model the solidified material, the concept of “metaballs” can be introduced [148] as implicit surfaces, which are created from a set of n particles and can be defined by the level set of a density function $\mu(x)$, as follows:

$$\mu(x) = \sum_{i=1}^n \frac{r_i^2}{\mathbf{x} - \mathbf{p}_{iL^2}^2} \quad (91)$$

where $\mathbf{p}_i$ represents the centroid of particle i with radius r_i . In order to avoid the motion of far-away particles to affect the local geometry, the density function can be modified with a finite support radius as follows [148]:

$$\mu(\mathbf{x}, r_{max}) = \sum_{i=1}^n \begin{cases} \mu(\mathbf{x}), & \mathbf{x} - \mathbf{p}_{iL^2} \leq r_{max} \\ 0, & \text{otherwise} \end{cases} \quad (92)$$

The concept of metaballs can be useful not only for the visualization of solidified parts of the material but also for the calculation of forces due to Marangoni effect. The force, which is exerted on a particle belonging to a metaball due to temperature gradients, is calculated as follows [148]:

$$f_{m_i} = -k \nabla(\mathbf{p}_i) T_i \quad (93)$$

where k is a scaling coefficient. This force can attract particles to the melt pool.

Other types of forces acting on the powder particles are gravity forces and viscous drag. These forces are considered as domain forces and are computed as follows [147]:

$$\mathbf{f}_{\Omega_i} = m_i \mathbf{g} - c_d \mathbf{v}_i \quad (94)$$

where m_i is the particle mass, $\mathbf{g}$ is the gravity acceleration and c_d is the viscous drag coefficient. When interactions between particles and rigid bodies are desired, planar spatial boundary conditions are imposed. These boundary conditions are implemented by moving or stationary planes which are not modeled using a detailed material model but their interaction is taken into account by additional normal and tangential forces on the particles, as follows [147]:

$$\mathbf{F}_{p n_{i,k}} = k_p (\mathbf{r}_i - \delta_b) \mathbf{n}_k \quad (95)$$

$$\mathbf{F}_{p s_{i,k}} = \mu_b f_{p n_{i,k} L^2} \bar{\mathbf{v}}_i \quad (96)$$

where k_p is a boundary stiffness parameter, δ_b the minimum distance between the particle i and the bounding plane k , μ_b the boundary friction coefficient and $\bar{\mathbf{v}}_i$ the projection of particle velocity on the boundary plane. As with the contact forces, the forces exerted by the boundary planes on a particle are also summed.

It was observed experimentally that denudation of the feedstock material occurs around and inside the melt pool region [148]. Due to the vaporization of feedstock material as well as residual moisture, a vapor plume is created, which causes the shield gas to develop a rapid flow locally [148]. Thus, particles from areas around the melt pool are displaced towards it. The force causing their movement can be modeled as follows:

$$f_{g p_i} = \frac{P}{(\mathbf{p}_{spot} - \mathbf{p}_i)^2} \quad (97)$$

where P is an experimentally derived pressure and $\mathbf{p}_{spot}$ represents the location of the laser heat source.

According to Haeri et al. [157], it is argued that a force due to the laser radiation pressure is exerted on powder particles. As the grain size in SLM/SLS is often at least one magnitude larger than the laser wavelength, the laser radiation pressure can be computed via the generalized Lorenz–Mie theory [157]. The detailed derivation of the force due to laser radiation pressure can be found in the work of Haeri et al. [157].

Apart from the mechanical forces, which lead to displacement of particles, the effect of heat on powder particles can

be also considered by DEM. When radiation and convection between particles can be neglected, the heat flux on a particle is calculated based on three different heat fluxes. Heat flux, which can occur on a particle i from its contact with a particle j , is calculated by [148]:

$$q_{c_{ij}} = k_{t_c} (T_j - T_i) \quad (98)$$

where k_{t_c} is the heat conduction coefficient which takes into consideration the properties and size of the powder particles and their contact cross-sectional area. Moreover, the particle bonds also conduct heat as follows [148]:

$$q_{b_{ij}} = k_{t_b} (T_j - T_i) \quad (99)$$

where k_b is also a heat conduction coefficient regarding this type of heat flux. Finally, the interaction of a particle and a boundary plane can produce a heat flux [148]:

$$q_{p_{ij}} = k_{t_p} (\tau_e - T_i) \quad (100)$$

where τ_e is the temperature of the surrounding environment. However, when radiation and convection cannot be neglected, their contributions are computed in the same way as they are computed for continuum domains as follows [149]:

$$q_{CONV} = h_{CONV} (T_i - \tau_e) \quad (101)$$

$$q_{rad} = \varepsilon k_B (T_i^4 - \tau_e^4) \quad (102)$$

where h_{CONV} represents the convective heat transfer coefficient, ε is the material emissivity, k_B is the Stefan-Boltzmann constant, equal to $5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$. These terms are active for particles, which are exposed on the top surface of the powder bed.

The various types of heat flux are finally summed for each particle, as well as the contribution of external heat sources. In the case of SLM/SLS, the external heat source is the laser beam. One requirement for the calculation of the amount of energy transferred to the powder bed by the laser beam is the definition of the trajectory of the laser beam. When the trajectory is defined, the heat flux is calculated as follows:

$$q_{\gamma_i} = \begin{cases} f_R(\mathbf{p}_i - \mathbf{P}_{nL^2}), & 0 \leq \mathbf{p}_i - \mathbf{P}_{nL^2} \leq r_e \\ 0, & \text{otherwise} \end{cases} \quad (103)$$

where f_R represents a radial function, which describes the distribution of heat flux within the sphere, such as a uniform or a Gaussian distribution and $\mathbf{P}_n$ represents the position of the center of the laser beam each time step.

Phase change can be also taken into account, e.g. by the apparent heat capacity method, in which the amount of energy required for the phase change is considered through

an appropriate modification of the specific heat value of the material in a small range of temperatures around the melting or vaporization temperature [149].

Regarding the modeling of heat transfer of powder beds with DEM, it is often considered an accurate approach as the problems with the powder thermal conductivity calculations, which affect the heat transfer calculations considerably, seem to be adequately solved. However, some authors point out that neglecting the modeling of the effect of gas in the powder bed as a second phase, leads to underestimation of heat transfer in powder beds, as the calculations are based on point-to-point contacts [158].

Finally, the coupled thermo-mechanical model can be solved by numerical integration. At first, the differential equation of the mechanical part of the model is solved [147]. A system of coupled differential equations is solved for the translation motion as follows:

$$\frac{d}{dt} m_i \mathbf{v}_i = \mathbf{f}_i \quad (104)$$

$$\frac{d}{dt} \mathbf{p}_i = \mathbf{v}_i \quad (105)$$

The first equation of the system is essentially the second Newton's law considering all the aforementioned force contributions, which can be expanded as follows:

$$\frac{d}{dt} m_i \mathbf{v}_i = \mathbf{f}_i = \mathbf{f}_{\Omega_i} + \mathbf{f}_{c_i} + \mathbf{f}_{b_i} + \mathbf{f}_{p_i} \quad (106)$$

It is to be noted that in the above equation only the most usually employed terms are included but however, depending on the case, additional force terms can be also added. The velocity of the particles $\mathbf{v}_i$ can be obtained by integrating the previous equation by velocity-Verlet scheme:

$$\mathbf{v}_i^{t+\Delta t} \approx \mathbf{v}_i^t + \frac{1}{2m_i} (\mathbf{f}_i^t + \mathbf{f}_i^{t+\Delta t}) \Delta t \quad (107)$$

Finally, the position vector of each particle can be obtained by integrating the second equation of the system:

$$\mathbf{p}_i^{t+\Delta t} = \mathbf{p}_i^t + \mathbf{v}_i^t \Delta t + \frac{1}{2m_i} \mathbf{f}_i^t \Delta t^2 \quad (108)$$

Accordingly, for the rotational motion of the particles, another system of differential equations is solved [159]:

$$\frac{d}{dt} \boldsymbol{\omega}_i = \mathbf{I}^{-1} \boldsymbol{\tau}_i \quad (109)$$

$$\frac{d}{dt} \boldsymbol{\theta}_i = \boldsymbol{\omega}_i \quad (110)$$

where $\boldsymbol{\theta}_i$ is the orientation of the particle, $\boldsymbol{\omega}_i$ is the rotational velocity, $\mathbf{I}_i$ is the inertia tensor of the particle and $\boldsymbol{\tau}_i$ is the

total torque. If the rotation is calculated in the global coordinate system, calculations for individual particles by the equation regarding ω_i should include multiplication by the rotation matrix R_i in the right side of the equation [145]. The inertia tensor for spherical particles can be calculated as follows:

$$I_i = \frac{2}{5} m_i \left(\frac{1}{2} d_i \right)^2 \mathbf{I} \quad (111)$$

where $\mathbf{I}$ is the identity matrix.

For the heat transfer problem, the following equation is required to be solved [147]:

$$\nabla^2 T_i = q_i \quad (112)$$

In order to solve the previous equation, various schemes can be adopted such as the implicit Euler method:

$$T_i^{t+\Delta t} = T_i^t + \frac{q_i^t}{m_i} \Delta t \quad (113)$$

The value of Δt is restricted in order to ensure stability of the solution process. The critical time step is defined in terms of particle system stiffness and mass as follows:

$$\Delta t_{crit} \cong \sqrt{\frac{m_{min}}{k_n}} \quad (114)$$

Another definition of the critical time step is the Rayleigh critical time step [153] which takes into account more parameters of the model:

$$\Delta t_c = \pi \left[\frac{R_i}{0.163 \xi_i + 0.877} \sqrt{\frac{2 \rho_i (1 + \xi_i)}{E_i}} \right]_{min} \quad (115)$$

Depending on the characteristics of the simulations, the critical time step may be reduced even to the 20% of its original values for reasons of stability [153]. In general, apart from the purely numerical parameters of the DEM method, which should be chosen after appropriate investigations in order to ensure stability and accuracy of the simulation results, several researchers also carry out investigations regarding other parameters whose values cannot be directly measured, such as the friction coefficient, restitution coefficient or surface energy density [154].

8.2 Modeling of the Powder Bed Setup for SLM/SLS Simulations

After the fundamentals of DEM were discussed, the representation of the powder by discrete particles, as well as the spreading process used for the powder particles carried out by DEM will be presented.

For the representation of a powder bed, two fundamental aspects exist [151, 160]: the representation of particle size and powder bed particle distributions. Regarding particle size, its distribution can be determined either by artificial methods, by using a statistical distribution such as uniform, bipolar or normal or by experimental methods. When artificial methods are selected, in each case a different assumption is made; for uniform distribution the powder particles have a single diameter, for bipolar distributions two different particle diameters can be considered, whereas for the normal distribution, particle diameter is governed by a Gaussian distribution curve [160]. In the case where experimentally determined distribution can be obtained, a more realistic size distribution of the particles is expected and common methods include the use of a laser particle sizer for the calculation of particle diameters based on laser diffraction [161, 162] or the analysis of SEM micrographs of the powder [145, 163, 164]. Then, statistical methods are applied to obtain the distribution curve by fitting the experimental data to a correlation function, e.g. normal distribution.

After the determination of the particle size distribution, the laying distribution of the powder particles in the powder bed should be determined, e.g. the particle locations and radii [160]. Various methods are proposed in the relevant literature, such as regular distribution, raindrop model (RDM) and simulation by DEM [151, 160]. When the regular distribution method is adopted, artificially designated particle locations are required and the powder bed is created based on the particle size distribution. This method is suitable mainly for simplified calculations, as it represents an ideal distribution of the powder bed particles. In the raindrop model, the new particles are first fixed in the vertical direction and then rotated to reach their equilibrium positions. A problem arising in the case of RDM is that the packing fraction cannot be controlled and it is required that some particles are added or deleted in order to obtain an accurate packing fraction [151]. Finally, DEM method, which was thoroughly presented in a previous subsection, leads to a more realistic particle distribution of the powder bed but requires considerably larger computational cost than the two previous methods.

When DEM is used, usually the fundamental parts of powder spreading setup are directly modeled. Two main types of spreading are modeled, the one involving a spreader blade, termed also as brush or wiper and the second involving a roller [146, 151, 162]. In both cases, the first step of the simulation consists of the creation of a random cloud of powder particles, able to drop freely due to the effect of gravity [144, 160, 165]. In some cases [165], the random cloud can follow a specific distribution according to real measurements. e.g. based on SEM micrographs. Rarely, the poured packing method [145] is employed for the initial powder bed distribution or also some advanced, dedicated

deposition algorithms [159]. Finally, a “raining” method was also described, during which the particles are initially placed in a horizontal plane, fall downwards towards a bounded box and then the side walls of the box are removed [166]. This method achieves the lower average packing fraction, compared to the random closed packing or the hexagonal crystalline packing [166].

When the particles reach the lower surface, where they are placed before the blade or roller pushes them towards their final positions, they are allowed to relax and settle for a time period before the spreading process begins. The powder particles can be initially placed on top of a container [144], a feedstock plate [162] or even a special delivery chamber, with a certain depth in order to be able to store a sufficient quantity of powder to be used later, as in real SLM/SLS processes [153]. Usually, a small part of the real setup, up to a few mm is modeled, with powder bed sizes of a few hundred μm , in order to achieve reduced computationally cost, and periodic boundary conditions are placed in the normal direction to the plane where the blade movement occurs [167]. In the cases where a spreader blade is used, the blade is moving horizontally at a constant speed towards the particles [144]. In most cases in the relevant literature, the blade is placed in a position perpendicular to the bottom plane of the spreading setup and the powder bed and its height is adjusted according to the geometry of the spreading setup.

Various powder spreading setups have been employed in previous works, with most of them having only subtle differences between them. In the simplest case, where the initial powder particles are deposited in one part of a long plate and subsequently moved to their final location, the blade should be positioned at such a height that the gap between it and the plate is almost equal to the desired thickness of the powder bed. In [144], the initial container of the powder particles was higher than the powder bed and the displaced particles were falling off a step in order to enter the powder bed, without the need of lifting the blade at suitable height. A similar setup is represented in [162, 165], where a special building chamber is intended to store the powder bed particles. In these works, the initial amount of powder was larger than the one required and thus, when the building chamber was filled, the rest of particles were removed. Finally, the different regions can be separated by a wall and in this case the blade is situated exactly above the wall height [153]. Moreover, the blade geometry plays an important role on the development of the powder bed and thus, apart from the simple straight geometry, some researchers have also investigated the efficiency of chamfered blades [151] or blades with special curvature, derived from super-elliptic profiles [164].

The use of a roller is an alternative powder spreading method in order to create a powder bed with the desired properties. Rollers move at a constant feed and rotate with a constant rotational speed. An important parameter of the

roller is the direction of rotation, which in the majority of works is counterclockwise. The importance of counterclockwise rotation was determined in [144], as it was shown that a clockwise rotation leads to a backward movement of the packed powder particles and finally, the thickness of the powder bed is higher. The rotation speed affects directly the solid ratio of the powder bed and it is suggested that it should be kept in relatively low values [144]. Although in most cases, the roller and base plate is assumed flat, Nan et al. [145] modeled both of them as rough surfaces, based on direct measurements or their surface roughness. The roller is often separated from the base plate by a gap equal to the desired height of the powder bed.

Finally, after the powder spreading simulation is over, the geometry of the powder bed can be exported in suitable file format in order to be used by the FEM code [165].

9 Meshless Approaches for SLM/SLS Processes

Apart from Discrete Element Method, which was discussed in the subsection regarding the modeling aspects of powder in SLM/SLS processes, there are also a few studies conducted with the aid of other meshless methods such as the Smooth Particle Hydrodynamics, Optimal Transportation Meshfree method, as well as Molecular Dynamics, which is especially suited for nanoscale analysis of additive manufacturing processes. These methods are used particularly due to their capability of handling large deformations occurring due to the intense thermal phenomena in the powder and substrate regions and their ability to capture free surfaces or interfaces without the need of additional approaches such as volume of fluid, level set or phase field methods. [168]

9.1 Continuum Level Meshless Approaches

One more modeling and simulation approach of SLM process is by implementing the meshless method of Smoothed Particle Hydrodynamics (SPH). In SPH the domain is presented by a set of particles, having individual material properties, which move according to governing conservation equations. In contrast with traditional grid-based numerical methods, SPH owns some special advantages. It is an inherent adaptive modeling technique, since the field variable approximation is performed at each time step based on a current local set of arbitrary distributed particles, resulting the adaptability of the method. Due to that adaptive nature of SPH approximation, the formulation in SPH is not affected by the arbitrariness of the particle distribution, a feature extremely favorable in modeling of powder bed processes, where particles have an arbitrary distribution. In brief, SPH is suited to handle and model complex flows,

with free surfaces, large deformations, moving interfaces, and materials splitting [169, 170]. In the study of Shah and Volkov [171] a precursor work regarding the utilization of SPH in modeling SLM/SLS is found. They presented an SPH based numerical method to simulate the heat transfer in a 2D powder bed system, using finite thermal contact conductance at interfaces between powder particles. Their model could feasibly be generalized for 3D simulations, as well as for systems composed of non-spherical particles. Russell et al. [170] employed the SPH method to simulate laser fusion additive manufacturing processes, including SLM. As in FEM models, Navier–Stokes equations, the surface tension and Marangoni convection were modeled, along with temperature dependent material properties. Moreover, the thermal model includes a Gaussian 3D volumetric heat source, with the power density across the powder bed thickness being defined based on the Beer-Lambert law. The mathematical expression of physics is almost common with FEM method, whereas differentiates mainly in the numerical implementation and solving. The computational results include the temperature profile, velocity and surface tension fields and the material state, meaning solid, liquid and in-between states. Weirather et al. [172] presented a 3D multiphase, weakly compressible SPH model to simulate SLM process. Viscous forces, surface tension, thermo-capillary forces, conductive heat transfer, melting and solidification were considered, whereas, the evaporation effect, i.e. developed recoil pressure, was incorporated within the SPH framework. In addition to the temperature fields, the length, width and depth of melt pool for a single track and different machining parameters were calculated, while, simulation results compared with experimental ones. Furthermore, studies regarding the influence of surface tension and the morphological characteristics of the molten pool when using SPH method have been conducted [173–175].

Bierwisch [176] presented an interesting model, which couples the Discrete Element Method (DEM) with SPH, where DEM was applied to simulate the powder spreading process and SPH was utilized to solve the melt pool dynamics. In DEM, the particles interaction was calculated by considering Hertzian repulsion, viscous damping, cohesion, sliding friction and rolling friction of the powder particles, while in SPH, the thermal and fluid dynamic features were incorporated. Temperature and velocities fields, along with some morphological characteristics of the molten pool, like its height profile, were obtained [176].

In sake of completeness the Optimal Transportation Meshfree (OTM) is referred. OTM classified as a mesh-free Galerkin method, with the primary variables and their time derivatives being nodal quantities. Mass, volume, density, stress and heat conduction are defined at material points, while during the computation, a search algorithm establishes the connectivity between material points and nodes. OTM is

a relatively new method, since it was initially introduced in 2010 by Li et al. [177] and although it has not been extensively used in modeling of SLM/SLS, it seems promising. Namely, Wessels et al. [178] used OTM to study the melting and solidification of laser irradiated particles during SLM, incorporating the relevant thermal and fluid dynamic physical phenomena. In a later work, Wessels et al. [179] presented a comparison study between the implementation of a volumetric heat source, and the use of ray tracing technique for defining the spatial laser power distribution. Finally, under a more generic approach, Fan and Li [180], proposed the utilization of OTM for modeling and simulation of AM processes of metals.

It has to be pointed out that details regarding the numerical solving and the computational framework of the aforementioned methods are intentionally not reviewed. Any reference would be superficial, lacking the necessary in depth analysis.

9.2 Molecular Dynamics

Another meshless approach for modeling aspects of the SLM/SLS processes involves the use of Molecular Dynamics (MD) method. This method is applied to systems of atoms and can accurately model their interactions in order to predict various phenomena, which can be properly explained only by taking into consideration the atomistic structure of materials. Using MD method, the interactions of atoms are calculated through suitable potential functions and atomistic trajectories are derived by numerical solution of Newton's equations of motion for each atom in the system.

Although it can be considered as a meshless method, the difference from most other meshless methods is that this particular one does not solve continuum equations such as SPH. Moreover it is by no means possible to be used for the simulation of the entire process of SLM/SLS or even their micro-scale counterparts, but it can be used for detailed study of phenomena occurring during SLM/SLS processes, such as the formation and growth of crystal nuclei and epitaxial grain growth, which can only be examined in atomistic scale. More specifically, given the relatively high computational cost of the MD simulations, the size of the computational models is restricted and is usually much lower than 1 μm . More specifically, the dimensions of models in most cases lie in the range of tens of nanometers, whereas the larger model has dimensions of $3438.25 \times 1340 \times 57.2088$ Å with a total number of 14,050,000 atoms [181]. In comparison to the nano-machining simulations, this value is relatively high as common simulations do not exceed a few hundred thousand atoms and the maximum number of atoms is about a few tens of millions. The diameter of powder grains in the SLM models lies usually in the range of 2.2–10.8 nm [181, 182]. With this method, only a few works have been

conducted and some representative works have been selected to be presented, mainly pertinent to the study of melting and solidification process of the powder as well as the determination of appropriate conditions to avoid crystallization of parts created from amorphous alloy powders fabricated via SLM processes.

Kurian and Mirzaeifar [181] investigated the phenomena of formation and growth of crystals during micro-SLM process of aluminum nano-powder with a 2D MD model. Their model included three layers of aluminum nano-powder particles of different sizes, which were heated by a laser beam in multiple passes. Their detailed observation of the evolution of the structure of the powder bed allowed them to determine the effects of melting and cooling stages of the SLM process. The heat in the powder bed was sufficient to fill the voids and create a homogeneous melt pool and then, during the cooling process the growth of new grains from the molten nano-powder took place, firstly epitaxially along the directions of maximum temperature gradient and afterwards by a homogeneous process of nucleation at the center of the melt pool, where the cooling rate was higher. Various defects and microstructure alterations were observed as well as the effect of each preceding track to the new ones. Finally, it was shown that it is possible to remove the nano-pores from the solidified material by hot isostatic pressing almost achieving full density.

Nandy et al. [183] studied the process of consolidation of metal nano-powders during laser sintering using an MD model. Two different models, consisting of two nanoparticles each, were created; in one of the models, the two particles are of the same size and in the other the diameter of one particle is twice the diameter of the second one. Their results indicated that melting conditions on the outer surface of the nanoparticle affect the neck formation between the two particles and that, increased temperature leads to the growth of the neck. Moreover, shrinkage at the early stages of sintering is promoted in the case with equal-sized particles as higher surface diffusion occurs whereas at the later stages, higher shrinkage occurs in the case of unequal sized particles. In another study, Nandy et al. [184] compared the behavior of even and uneven nano-particles of different sizes and shapes during sintering with five different laser density values. It was found that sintering rate is independent of the particle size and increases with increasing laser energy density, that unevenly sized particles exhibit more intense surface and volume bulk diffusion and higher shrinkage ratio, whereas cubic particles and hollow cylindrical particles exhibited the lowest and highest rate of diffusion, respectively.

Regarding nanoparticle laser sintering, Pan et al. [182] also investigated the mechanisms governing neck growth in the nanoscale. Nanoparticles of three different sizes were employed, with their diameter varying from 2.2 to 10.0 nm. For each diameter, the model included two equal sized

nano-particles which were subjected to irradiation from a continuous laser beam after an initial neck began to form between them. It was found that the neck growth mechanisms varied with different size of nanoparticles, as grain-boundary slidings and surface diffusion occurred for the case with the lower diameter, grain-boundary diffusion, surface diffusion and viscous flow occurred for the intermediate size particles whereas for the largest nanoparticles grain-boundary diffusion and surface diffusion occurred. The MD model was superior to an analytical one, which exhibited similar results only for the case with the largest particles. Tan et al. [185] developed an MD model with a view to determine the optimum conditions, which lead to lower contamination of the unused powder, in cases with powder beds with different powder materials. Two models of the powder bed were used; in the first model, iron and aluminum particles of 5 nm diameter were aligned in two adjacent rows and in the second one, the particles were arranged in two layers of four rows of particles. After laser irradiation, the melting process was studied.

Other researchers used MD models in order to investigate the optimum conditions for avoiding the crystallization of parts fabricated by amorphous alloys. For example, Jiang et al. [186] conducted simulations on SLM processing of Fe₅₀Ni₅₀ amorphous alloy with a view to determine the optimum scanning speed and laser power which reduces crystallization. Their model included both a powder bed consisting of nanoparticles and a substrate. They found that low scanning speeds and high laser power values favor the creation of BCC bond pairs whereas lower relaxation time leads to a considerable decrease of the creation of BCC and other crystal types. Finally, Guo et al. [187] conducted a similar study regarding the SLM process of Ni₇₅Al₂₅ under different heating and cooling rates. They found that higher cooling rates reduce considerably or even eliminate the nucleation of crystals after a critical value.

10 Summary

As SLM and SLS continuously gain ground in modern industrial environment, the need for a deeper study and better understanding of the processes arises, with modeling and simulation being a powerful tool in research. Nevertheless, due to the inherent complexity of the processes, since multi-physics phenomena occur and as additive manufacturing is a novel technology, modeling of SLM/SLS is still a scientifically open topic. Different modeling methods and strategies have been presented and proposed, each one having its own strengths and weaknesses, aiming to the most accurate and realistic simulation results, considering the existent limitations in computational power. In the current paper a comprehensive overview on the different modeling

methods, assumptions and simplifications, through which researchers mathematically express and model the underlying physical mechanisms, were presented. The topic was covered by analyzing each sub-model that can be implemented on a simulation.

Specifically heat transfer consists the basis of every modeling and simulation, while fluid dynamics is often employed in a multiphysics coupled simulation. Parameters, like the absorption coefficient, the spatial distribution of the laser beam power, the powder bed thermophysical properties and boundary conditions, e.g. convection coefficient and emissivity, can be estimated according to experimental data, semi-empirical relations, numerical, or analytical solutions. Additionally, residual stresses, the microstructure evolution and, in SLS, the crystallization development, are estimated by incorporating the proper sub-models. The obtained data depend on the implemented models, based on the principal research aim. Furthermore, in the current review, modeling methods pertaining to the powder distribution and powder bed set-up were presented. The mainly used method is DEM, a Lagrangian approach in modeling granular mediums, while the results can be subsequently used in a FEM model. Finally, other meshless methods, i.e. SPH and Molecular Dynamics were reviewed, as they have been proposed as alternatives of FEM in simulation of SLM and SLS.

As general and main conclusion, it can be worded that modeling SLM is not a straight forward procedure, as in the core of simulation lies a complex and multi-parameter process. There is no uniform and general modeling framework, rather than developed sub models that have to be incorporated in an efficient way, forming a realistic, accurate and, in terms of computational time, feasible simulation. Researchers and scientists have to be aware of the available methods and act on a case-by-case basis, following of course some general rules and guidelines. With passing time, a more robust and complete modeling framework will be developed, utilizing the keep rising available computational power. A better and more detailed insight of the SLM/SLS will advance its development, resulting to render it as feasible and competitive manufacturing alternative process.

Declaration

Conflict of Interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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